Beyond Specific Degradations: Prompt-Driven Spatial–Spectral Modeling for Hyperspectral Image Enhancement

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***Abstract*—Hyperspectral images (HSIs) are often degraded by complex and coupled corruptions such as noise, blur, and haze. Most existing methods are designed for specific degradation types, lacking the flexibility for unified enhancement. In addition, jointly modeling spatial structures and spectral correlations remains challenging. In this article, we propose a prompt-driven spatial–spectral framework for HSI enhancement, which improves adaptability across diverse degradations via semantic–visual–spectral (SVS) multiprompt learning. The proposed framework introduces cross-modal prompts to provide adaptive guidance, enabling the model to handle diverse, mixed, and shifted degradation patterns within the considered HSI degradation space. To extract richer spatial representations, we design a direction-aware rectangle self-attention (DRSA) mechanism. It effectively captures horizontal and vertical dependencies to better restore anisotropic structures while reducing computational cost compared with dense global self-attention. To preserve spectral fidelity, we introduce a frequency-aware prompt-guided spectral modulation (FPSM) mechanism, which dynamically modulates spectral features using a learnable spectral prototype bank. We further devise a cross-modal semantic–visual dynamic prompting (SVDP) mechanism that integrates textual degradation descriptions and visual prototypes to generate degradation-aware prompts, enabling adaptive perception of diverse corruption patterns. Extensive experiments on remote sensing HSI datasets show that the proposed method achieves performance superior to state-of-the-art methods. It consistently improves both visual quality and spectral accuracy under various degradation conditions. These results validate the potential of multiprompt learning for unified HSI enhancement tasks.**

Received 24 June 2026; revised 27 July 2026 and 1 September 2026; accepted 22 September 2026. Date of publication 30 September 2026; date of current version 6 October 2026. This work was supported in part by the National Key Research and Development Program of China under Grant 2024YFB2907700, in part by the National Natural Science Foundation of China under Grant 62571181 and Grant 62271192, in part by China Postdoctoral Science Foundation under Grant 2025M781634, in part by the Program for Science and Technology Innovation Talents in Universities of Henan Province under Grant 26HASTIT011, in part by the Natural Science Foundation of Henan under Grant 262300421299, in part by Henan Provincial Key Research and Development Program under Grant 261111242400, in part by the Training Program for Young Backbone Teachers in Higher Education Institutions of Henan Province under Grant 2025GGJS040, and in part by the Scientific and Technological Project of Henan Province under Grant 262102211006. *(Corresponding author: Pengju Si.)*

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Digital Object Identifier 10.1109/TGRS.2026.3738062

***Index Terms*—Direction-aware attention, hyperspectral images (HSIs) enhancement, multiprompt learning, spectral fidelity, unified enhancement framework.**

I. Introduction

HYPERSPECTRAL images (HSIs), structured as high-dimensional data cubes, provide dense spectral signatures in hundreds of continuous narrow bands. This rich spectral information enables fine-grained analysis in remote sensing, such as land-cover classification [1] and environmental monitoring [2]. However, HSIs acquired through airborne and satellite platforms are frequently compromised by composite degradations, such as noise [3], haze [4], and spatial-resolution degradation [5], as shown in Fig. 1(a). These degradations are typically coupled in both the spatial and spectral domains, causing severe structural distortion and spectral inconsistency, which greatly increases the difficulty of image enhancement.

Existing HSI enhancement methods are mainly designed for specific degradation types [4], [5], [6]. Although they achieve promising performance in controlled settings, their generalization ability is limited when facing mixed, or complicated degradation conditions in real-world scenarios. In addition, HSIs exhibit strong spatial self-similarity [7] and high spectral correlation, requiring enhancement models to jointly capture nonlocal spatial dependencies [8] and cross-band interactions. Therefore, developing a unified framework that can adapt to diverse degradations while preserving spectral fidelity remains highly challenging. Meanwhile, complex mixed degradations further hinder effective spatial–spectral representation learning, leading to insufficient structural recovery and spectral distortion.

Recent advances in deep learning [9] have significantly improved HSI enhancement, particularly in spatial–spectral representation learning. Existing methods often rely on localized modeling [10], which is effective for capturing fine-grained textures but inherently limited by restricted receptive fields, leading to insufficient aggregation of nonlocal spatial information. To address this limitation, global modeling methods based on self-attention [5], [11], [12] have been introduced to capture long-range interactions. However, their quadratic computational complexity with respect to spatial resolution poses a significant scalability challenge for high-dimensional

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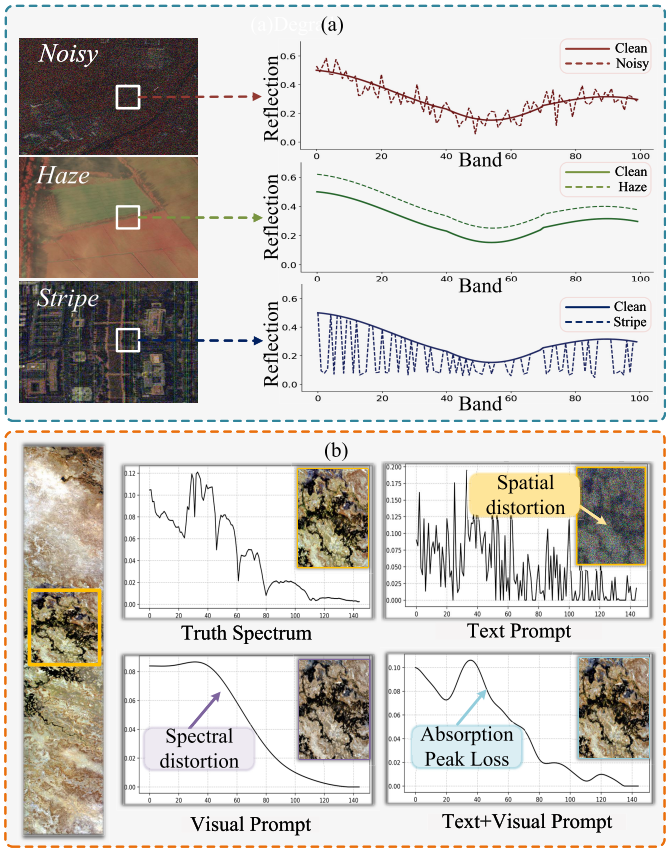

Fig. 1. Challenges of HSI enhancement and motivation of multiprompt learning. (a) Effects of different degradation types (noise, haze, and stripe) on spatial structures and spectral signatures of HSIs. (b) Enhancement performance under different prompt settings, illustrating how text and visual prompts influence spectral fidelity and structural recovery.

hyperspectral data. More recent paradigms, such as diffusion-based methods [13], [14], attempt to model the enhancement process from a probabilistic perspective by learning the underlying data distribution through progressive denoising. While this formulation provides strong generative flexibility, it relies on iterative sampling and stochastic reconstruction, which may introduce instability across spectral bands and lead to high inference cost. In parallel, state space methods, such as Mamba [15], [16], reformulate spatial modeling as a sequence modeling problem with linear computational complexity. These methods emphasize efficient long-range dependency modeling via selective state transitions. However, their sequential token-wise processing lacks explicit spatial inductive bias, making it difficult to accurately model structured spatial patterns in 2-D scenes. Therefore, designing an efficient and direction-aware spatial modeling mechanism remains a critical challenge for HSI enhancement.

In addition to spatial modeling challenges, handling diverse types of degradation remains a formidable challenge. In the RGB domain, all-in-one enhancement has been explored using prompt-based learning [17], [18], [19], [20], [21], [22], [23]. As shown in Fig. 1(b), visual prompts [20], [21] can guide feature modulation but often lack interpretability. Textual prompts [22], [23] provide semantic descriptions of degradations, but they may not align well with hyperspectral data. This mismatch is caused by the domain gap between natural images and hyperspectral imagery. Directly transferring vision-language models [17], [18] may introduce spectral distortion or artifacts. As a result, existing prompt-based methods cannot provide reliable guidance for hyperspectral enhancement. Furthermore, hyperspectral enhancement requires accurate modeling of spectral structures. However, many existing methods lack explicit prior spectral guidance. They often treat all spectral bands equally and ignore the intrinsic low-rank properties [3], [24] and interband correlations of hyperspectral data. This limitation leads to suboptimal spectral reconstruction, especially under complex or severe degradations.

In summary, despite recent progress, HSI enhancement still faces three key challenges as follows.

1) Simplistic Degradation Assumptions: Most existing methods focus on single or predefined degradation types. They neglect complex corruptions within the considered HSI degradation space that co-occur in real-world scenarios. This leads to poor generalization, increased storage cost, and reduced flexibility when relying on task-specific paradigms.

2) Inefficient Spatial–Spectral Modeling: Current architectures prioritize either local receptive fields or standard square-window attention. They overlook the anisotropic nature of hyperspectral data, such as elongated structures and directional stripe noise. Consequently, they struggle to capture long-range spatial dependencies while preserving spectral continuity, resulting in broken textures or blocking artifacts.

3) Lack of Refined Prior Guidance: Existing frameworks treat all spatial regions and spectral bands equally. They ignore the need for degradation-aware, multimodal priors that adapt to varying corruption patterns. This omission causes an imbalance between spatial structure recovery and spectral fidelity, often introducing spectral distortion or detail loss in severely corrupted areas.

To address these challenges, we propose a prompt-driven spatial–spectral modeling framework for HSI enhancement through semantic–visual–spectral (SVS) multiprompt learning. The framework explicitly integrates spatial structure modeling, spectral representation learning, and degradation-aware prior guidance within a unified architecture. For spatial modeling, we design a direction-aware attention mechanism that performs self-attention along horizontal and vertical rectangular regions. This allows the model to better capture anisotropic structures and improves the restoration of directional textures. For spectral modeling, we introduce a frequency-aware modulation scheme that decomposes spectral features into multiple groups and performs local spectral attention within each group. By incorporating spectral prompts, the model adaptively characterizes the responses of different frequency components, enabling more effective suppression of noise-sensitive bands and enhancement of stable spectral information. To further improve adaptability, we develop a mechanism that integrates semantic descriptions and visual degradation patterns. The generated prompts provide complementary global guidance and are injected into the network to refine feature representations under varying degradation conditions.

The main contributions are summarized as follows.

1) *Domain-Specific Unified HSI Enhancement Framework:* We propose a unified degradation-handling paradigm

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within each spectral domain. Instead of training separate models for individual degradation types, the proposed framework proposes a single shared model to handle multiple degradations in the natural-scene or remote sensing HSI setting. This design reduces task-specific model redundancy while maintaining adaptability to diverse degradation patterns under a fixed spectral-domain configuration.

2) *Direction-Aware Rectangle Self-Attention (DRSA):* We propose a flexible rectangular attention mechanism that replaces traditional fixed-window designs. By enabling anisotropic receptive field adaptation, DRSA effectively captures structured spatial patterns, leading to enhanced modeling of directional features and nonlocal correlations.
3) *Synergistic SVS Prompting Mechanism:* We propose a synergistic prompting strategy that bridges the modality gap between RGB-pretrained vision-language models and hyperspectral data. By integrating spectral prompts with learnable spectral prototype priors and multimodal guidance, the proposed mechanism facilitates semantically aware degradation modeling while preserving high-fidelity spectral reconstruction.

Through these three innovations, our framework can effectively enhance the performance of HSI enhancement tasks, particularly demonstrating higher efficiency and robustness when dealing with various types of degradation and complex scenarios.

II. Related Work

A. HSI Enhancement

Early data-driven methods for HSI enhancement achieve significant improvements by learning direct mappings from degraded observations to clean images [10], [25]. However, their reliance on localized receptive fields inherently restricts their ability to capture nonlocal spatial dependencies and maintain global spectral consistency. To address this limitation, transformer-based architectures demonstrate strong capabilities in this domain. MST++ [26] introduces a multistage spectral-wise self-attention mechanism to explicitly model long-range spatial–spectral dependencies without destroying spectral fidelity. VolFormer [27] further broadens the representation capacity by exploring comprehensive voxel-level interactions across the 3-D hyperspectral cube. Nevertheless, these isotropic interaction mechanisms typically treat all spatial directions and spectral dimensions uniformly, overlooking the anisotropic nature of spatial structures and the frequency-varying responses of distinct spectral bands.

To address this issue, recent literature rapidly shifts toward advanced generative and sequential paradigms. Diffusion-based models, including UniRestore [13] and diffusion-driven unmixing frameworks [14], formulate enhancement as a progressive denoising process. Notably, LDERT [6] integrates latent diffusion with rectangle self-attention to mitigate the high inference costs of traditional diffusion processes while preserving structural details. Concurrently, state space models (SSMs) emerge as highly efficient alternatives. Alternatively, methods like MambaHSISR [15] and wavelet-enhanced Mamba networks [16] reformulate 3-D spatial–spectral interactions into 1-D token sequences to facilitate efficient processing. However, this forced 1-D flattening inherently disrupts the highly coupled spatial–spectral continuity of the original data cube. Furthermore, their rigid sequential scanning mechanisms lack explicit structural inductive biases, hindering the precise recovery of fine-grained spatial boundaries and local spectral consistency. Overall, existing methods still face challenges in balancing modeling capacity, computational efficiency, and structural awareness, especially under complex degradation conditions.

B. All-in-One Image Enhancement

Traditional enhancement models optimize distinct network weights for specific degradation types [4], [5], [6], prompting the development of all-in-one frameworks designed to handle diverse, co-occurring corruptions within a unified architecture. Existing all-in-one methods [6], [19], [22], [23], [28], [29], [30], [31], [32] can be broadly categorized into two paradigms. The first paradigm focuses on implicit degradation modeling, where a unified network is trained using diverse degradation data, often introducing contrastive or adversarial learning to improve robustness. While improving generalization to some extent, these methods lack explicit mechanisms to distinguish different degradation types. The second paradigm introduces degradation-aware guidance, where enhancement is conditioned on explicit degradation representations. Recently, prompt-based learning [17], [18], [19], [20], [21], [22], [23] has emerged as a highly flexible and effective realization of this explicit paradigm. To illustrate, InstructIR [23] employs human-written textual instructions to dynamically modulate the enhancement network, achieving high-fidelity results across diverse RGB degradation tasks. Alternatively, HAIR [19] employs hypernetworks combined with contrastive regularization to generate degradation-specific weights dynamically based on the input features. Within the hyperspectral domain, PromptIR [21] and PromptHSI [32] pioneer a unified framework by extracting task-specific visual prompts through a vision-language modulated frequency adaptation module to guide the enhancement. However, when deployed in complex real-world environments, these unified frameworks exhibit critical shortcomings. Most existing methods are developed in the RGB domain and primarily focus on spatial feature modulation. They experience severe domain gaps when transferred to HSIs, as they apply uniform modulation strategies that ignore the profound spatial–spectral coupling. Consequently, applying these models directly to mixed HSI degradations frequently induces severe spectral distortion and a critical loss of fine-grained spectral signatures.

C. Prompt Learning for Image Enhancement

Prompt learning, initially popularized in natural language processing, extends to computer vision through large-scale vision-language models such as CLIP [33]. It provides a flexible mechanism to incorporate external knowledge and

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adapt models to diverse enhancement tasks via additional conditioning signals without modifying the backbone architecture. Current state-of-the-art methods typically encode degradation priors using visual [20], [21], textual [22], [23], or multimodal prompts. For example, the Multimodal Prompt Perceiver [17] utilizes a perceiver-based architecture to adaptively fuse diverse prompt modalities, enhancing the generalization and fidelity of the enhancement process. Focusing on frequency-domain characteristics, AdaIR [20] extracts dynamic visual prompts to modulate frequency components, allowing the network to adapt smoothly to spatially varying degradations. Both visual and textual prompts struggle to accurately represent subtle band-wise variations and coupled spatial–spectral degradations. Furthermore, existing prompt-based frameworks [17], [18], [19], [20], [21], [22], [23] fundamentally ignore domain-specific HSI priors, such as interband correlations and low-rank spectral structures. This crucial omission inevitably leads to severe spectral distortion and the loss of fine-grained details under complex mixed degradations.

III. Method

In this section, we first present the problem formulation and motivation of the unified HSI enhancement task. We then present the overall architecture of the proposed prompt-driven spatial–spectral modeling framework for HSI enhancement. Finally, we detail three key components: a DRSA mechanism for spatial modeling, an FPSM mechanism for spectral modeling, and a cross-modal semantic–visual prompting mechanism (SVDP) for degradation-aware guidance.

A. Problem Formula and Motivation

Given a degraded observation Y and its corresponding latent clean image X, the HSI degradation process can be generally formulated as

$$Y = \mathcal{D}(X) + N \tag{1}$$

where $\mathcal{D}(\cdot)$ denotes a spatial–spectral degradation operator induced by the imaging system or environmental factors, and N represents additive noise with complex statistical properties. The operator $\mathcal{D}(\cdot)$ can take different forms under specific enhancement tasks. Typical examples include the following aspects.

1) Deblurring: In the deblurring scenario, $\mathcal{D}(\cdot)$ is commonly modeled as a spatial convolution operator

$$Y = K * X + N \tag{2}$$

where K denotes the blur kernel. This operation leads to spatial detail diffusion and may also affect spectral consistency.

2) Inpainting: In the inpainting scenario, the degradation process can be modeled as a binary masking operation

$$Y = M \odot X + N \tag{3}$$

where M is a binary mask matrix and $\odot$ denotes element-wise multiplication. This formulation simulates missing information caused by sensor defects or corrupted spatial regions.

In real-world scenarios, degradations are typically complicated and often occur in composite forms. The uncertainty of $\mathcal{D}(\cdot)$, together with the spectrally correlated noise, makes the inverse problem highly ill-posed. Therefore, a unified enhancement framework should be able to adaptively infer degradation-related representations under diverse degradation settings while jointly exploiting spatial texture priors and spectral correlation priors to achieve robust and high-fidelity hyperspectral reconstruction.

B. Overall Framework

Given a degraded hyperspectral observation $Y \in \mathbb{R}^{H\times W\times B}$, the network first employs a 3×3 convolutional embedding layer to extract shallow features $F_0 \in \mathbb{R}^{H\times W\times C_1}$, which retain essential spatial structures and the initial spectral distribution. A three-stage hierarchical encoder is then applied to progressively enlarge the receptive field via strided downsampling. The complete architectural pipeline, including specific spatial resolution reductions and channel expansions across the three stages, is illustrated in Fig. 2. This hierarchical design produces multiscale deep features denoted as F_n^e, where the superscript e indicates features extracted from the encoder, and the subscript $n \in \{1, 2, 3\}$ represents the stage index corresponding to different spatial resolutions. As n increases, F_n^e encodes progressively more abstract spatial–spectral representations. These multiscale features will be further utilized in the subsequent prompt generation process.

To facilitate adaptive enhancement under complex degradations, we introduce a global prompt generation mechanism. Specifically, the SVDP mechanism takes multiscale deep features F_n^e as input and produces a semantic prompt P_t, which serves as a global conditioning signal shared across the network. By encoding high-level degradation semantics, P_t not only provides global guidance for feature learning but also participates in skip-connected feature refinement during decoding via prompt-conditioned fusion, suppressing degradation-related responses and enhancing informative structures. To address the coupled degradation of spatial distortion and spectral inconsistency, we introduce an adaptive spatial–spectral block (ASB) at each stage of the encoder–decoder architecture. Each ASB operates on features with fixed spatial resolution and channel dimension at the corresponding stage. The ASB adopts a dual-branch design, where the spatial branch (DRSA) captures long-range spatial dependencies to restore structural details, and the spectral branch (FPSM) models cross-band correlations to preserve spectral fidelity. In this process, a learnable spectral prompt P_s is incorporated into the spectral branch to guide spectral feature modeling. The detailed designs of the DRSA, FPSM, and SVDP mechanisms are presented in Sections III-C–III-E.

C. DRSA Mechanism

HSIs exhibit pronounced spatial anisotropy, such as elongated object boundaries and directional stripe noise. Conventional square-window attention struggles to capture these patterns, often resulting in restricted receptive fields and blocking artifacts. To address this, we propose the DRSA mechanism. As shown in Fig. 3, DRSA replaces fixed square windows with parallel horizontal and vertical rectangular

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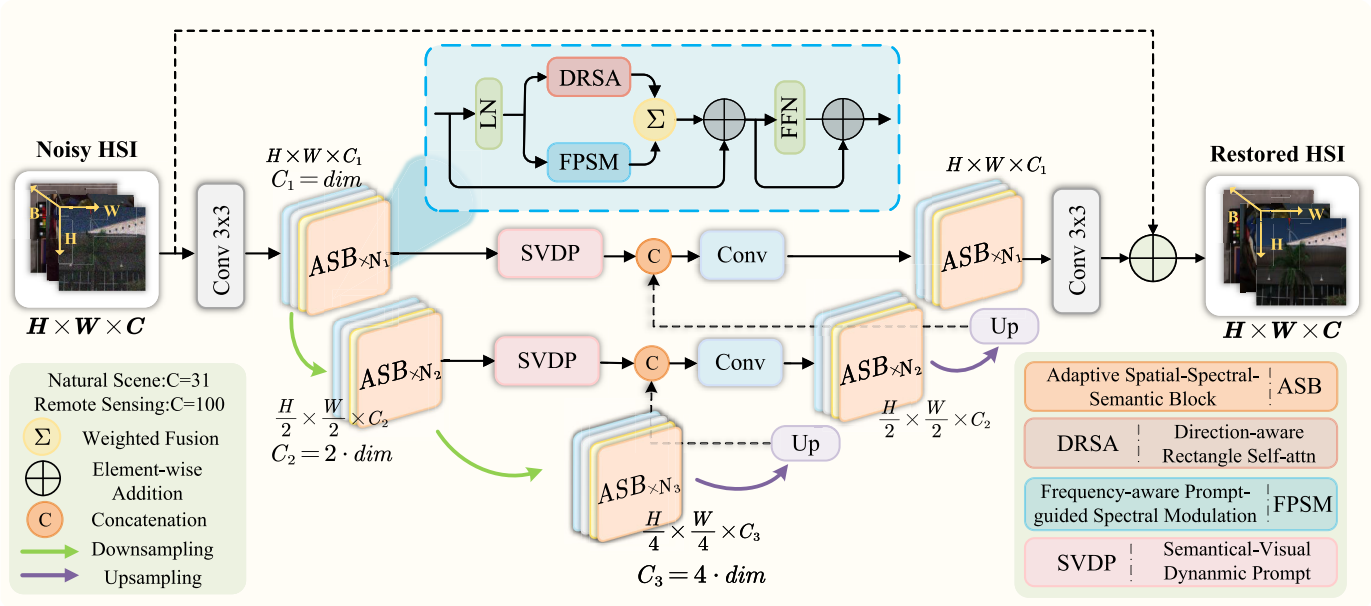

Fig. 2. Overview of the proposed prompt-driven spatial–spectral enhancement framework for HSIs. The model consists of semantic–visual dynamic prompting (SVDP), DRSA, and frequency-aware prompt-guided spectral modulation (FPSM). Here, C denotes the spectral dimension of the input–output pair under each spectral-domain setting, i.e., 31 bands for natural scene HSIs and 100 bands for remote sensing HSIs after preprocessing. The restoration model does not change the spectral dimension during enhancement.

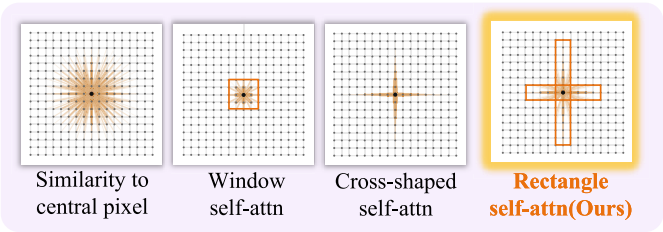

Fig. 3. Comparison of different spatial attention patterns.

partitions. This design introduces a directional inductive bias, expanding the axial receptive field to efficiently model long-range structural dependencies. Furthermore, to reduce computational overhead and enhance cross-dimensional interactions, DRSA adopts a paradigm of spectral splitting, spatial partitioning [34], and global rearrangement. Specifically, as illustrated in Fig. 4(a), given an input feature map $I \in \mathbb{R}^{H\times W\times C}$, we first divide it into two subfeatures along the spectral dimension

$$I^i \in \mathbb{R}^{H\times W\times \frac{C}{2}}, \quad i \in \{1,2\} \tag{4}$$

where each group is assigned to capture directional dependencies along a specific spatial axis. Following the stripe-based attention design, the feature maps are partitioned into rectangular stripes with predefined stage-specific widths. Specifically, for I^1, horizontal rectangular self-attention (H-MSA) is employed, where the feature map is divided into nonoverlapping horizontal stripes with size $[h, W]$. Here, h denotes the predefined horizontal stripe width. For I^2, vertical rectangular self-attention (V-MSA) is applied with stripe size $[H, w]$, where w represents the predefined vertical stripe width. The stripe widths are fixed for each network stage and are not adaptive to individual inputs. Different stripe widths are assigned across stages to balance local detail preservation and long-range dependency modeling. Smaller stripes are used in shallow stages to capture fine-grained structures, while larger stripes are adopted in deeper stages to aggregate broader contextual information. Compared with dense global spatial self-attention, which computes interactions among all spatial tokens, DRSA restricts the attention computation within directional stripe regions. This design reduces the computational cost while maintaining effective spatial dependency modeling.

To facilitate information exchange across neighboring windows, a shift operation is introduced prior to window partitioning. This operation allows tokens near window boundaries to interact with adjacent regions, thereby alleviating the discontinuity caused by independent window processing. Within each rectangular window, we further introduce an adaptive sparse modulation mechanism to refine the interactions captured by MSA. As illustrated in the expanded MSA structure in Fig. 4(a), the query–key responses are transformed into a dense attention map A_d through Softmax and a sparse response map A_s through squared ReLU.

The two branches provide complementary information for image enhancement. The dense attention map A_d maintains broad contextual interactions within each rectangular window, which is important for preserving structural continuity and recovering weak spatial details. However, dense attention may also propagate redundant or degradation-induced correlations because all token interactions participate in the aggregation. In contrast, the sparse response map A_s emphasizes reliable positive correlations and suppresses weak or negative responses, thereby providing selective cues for identifying informative interactions. Directly replacing A_d with A_s, however, may eliminate subtle but valid dependencies that are essential for

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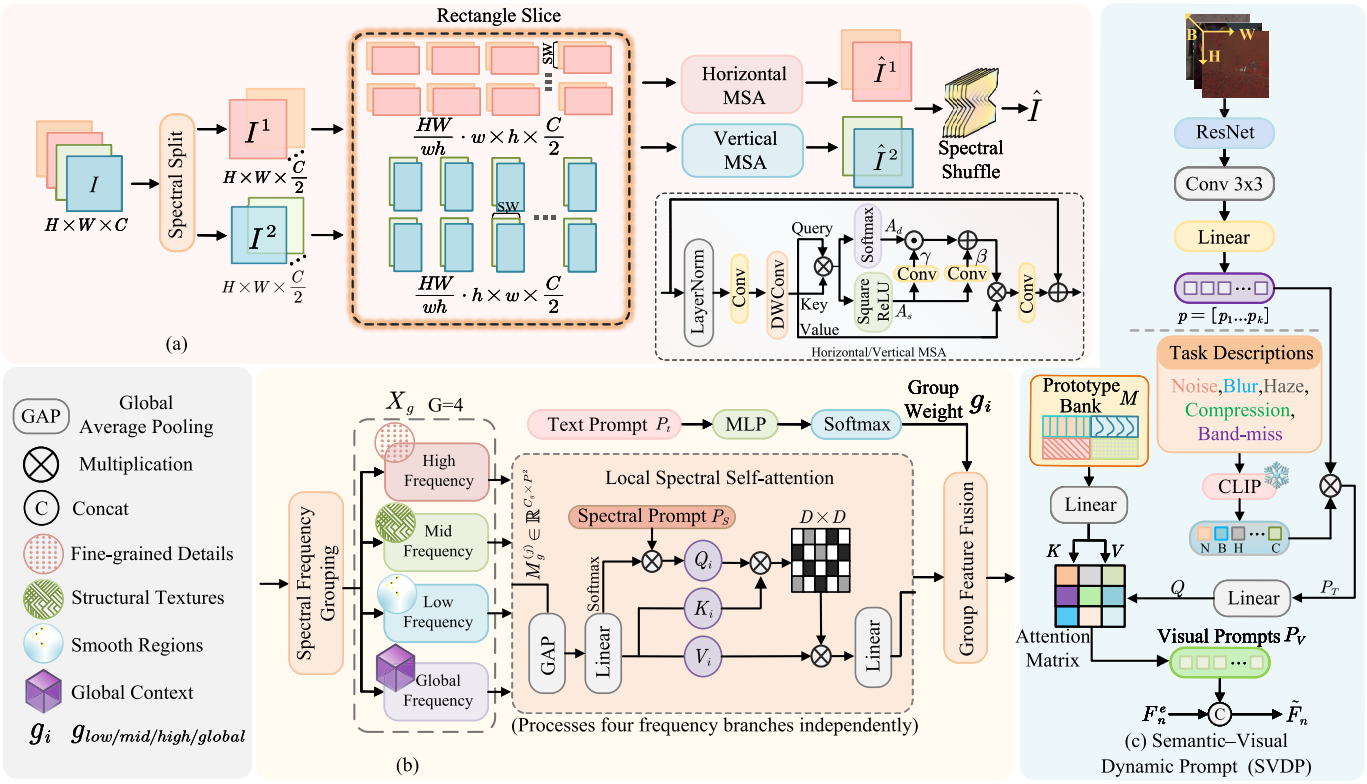

Fig. 4. Detailed design of the proposed key mechanisms. (a) DRSA, which captures anisotropic spatial dependencies via directional rectangular attention and spectral rearrangement. (b) FPSM, which decomposes spectral features into frequency groups and adaptively modulates them using spectral prompts, followed by a dynamic group feature fusion formulated as $X_{\text{out}} = \sum_i g_i \hat{X}_g$. (c) SVDP, which generates degradation-aware semantic and visual prompts via a cross-modal interaction mechanism.

detail reconstruction. Therefore, we deploy A_s to adaptively modulate, rather than replace, the dense attention map

$$\gamma = f_\gamma(A_s), \qquad \beta = f_\beta(A_s) \tag{5}$$

where $f_\gamma(\cdot)$ and $f_\beta(\cdot)$ denote lightweight convolutional projections, and γ and β represent the adaptive scale and shift factors, respectively. The final attention output is formulated as

$$\text{Attention}(Q, K, V) = (\gamma \odot A_d + \beta) V \tag{6}$$

where $\odot$ denotes element-wise multiplication.

In this formulation, γ performs content-dependent reweighting of the dense interactions, while β provides an adaptive offset to further refine the attention response. Consequently, the dense pathway retains sufficient contextual coverage and structural information, whereas the sparse pathway selectively suppresses unreliable and redundant correlations. Their cooperation prevents degradation-related responses from being indiscriminately propagated through dense aggregation while avoiding the loss of weak but meaningful image structures caused by purely sparse attention. This balance is particularly beneficial for image enhancement, where artifact suppression and detail preservation must be achieved simultaneously. Combined with shifted rectangular partitioning, the proposed mechanism improves both intrawindow interaction selectivity and cross-window structural continuity, enabling more effective recovery of boundaries, textures, and elongated spatial patterns. The V-MSA branch follows the same formulation, with the attention operation performed over vertically oriented rectangular windows.

The outputs from all rectangular windows are then aggregated to produce the final result of H-MSA

$$\hat{I}^1 = H\text{-}MSA\left(I^1\right) = Merge\left(I_1^1, I_2^1, \ldots, I_M^1\right) \tag{7}$$

where Merge(·) operation spatially concatenates the outputs of the M independent horizontal stripes back into a unified feature map of size $H \times W \times C/2$. Symmetrically, the vertical branch computes $\hat{I}^2 = V\text{-}MSA(I^2)$ to produce a vertically attended feature map with identical dimensions.

After extracting features along the horizontal and vertical directions, the resulting representations, $\hat{I}^1$ and $\hat{I}^2$, are first concatenated along the spectral dimension. The concatenated features are then further aggregated via a spectral rearrangement [35] operation. Unlike simple channel concatenation, spectral rearrangement disrupts the original channel order. This enables subsequent convolutional layers to fuse contextual information from different spatial directions. As a result, an effective global receptive field can be achieved. The final fused feature $\hat{I}$ is formulated as

$$\hat{I} = Rearrange\left(Concat\left(\hat{I}^1, \hat{I}^2\right)\right). \tag{8}$$

Consequently, DRSA explicitly encodes spatial anisotropy as an inductive bias, aligning its receptive field with dominant structural orientations. Meanwhile, the adaptive sparse modulation inside H-MSA and V-MSA reduces unreliable

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attention interactions, enabling more robust direction-aware spatial representation learning.

D. Frequency-Aware Prompt-Guided Spectral Mechanism

HSIs exhibit strong spectral correlations and degradation-dependent variations across bands. Since imaging conditions, atmospheric effects, and material reflectance characteristics vary with wavelength, different spectral responses may contain distinct spatial-frequency characteristics. This wavelength dependence motivates the model to exploit complementary spatial-frequency information for spectral restoration. Nevertheless, after convolutional projection and feature interaction, latent channels no longer maintain a one-to-one correspondence with the original wavelength bands. Therefore, FPSM does not directly partition latent channels into physically predefined spectral-frequency intervals; instead, it decomposes the learned latent representation into complementary spatial-frequency components and performs adaptive spectral interaction within each component.

To this end, we propose an FPSM mechanism, as illustrated in Fig. 4(b). Given an input feature $X \in \mathbb{R}^{H\times W\times C}$, a lightweight 1×1 convolution first projects its channel dimension from C to $C_g = C/G$, producing a compact latent feature while aggregating information across all input channels. We then employ a parameter-free residual decomposition to separate the projected feature into four complementary spatial-frequency-aware feature groups. Specifically, average pooling with kernel sizes of 3 and 5 is used to extract responses at different spatial scales, while global average pooling (GAP) captures global contextual information. Both average-pooling operations use a stride of 1 with appropriate padding to preserve the spatial resolution, while the globally pooled feature is broadcast to $H \times W$ before residual computation. Pixel-wise residuals between the projected and pooled features are progressively computed to obtain components $X_g \in \mathbb{R}^{H\times W\times C_g}$, which correspond to fine-grained details, structural textures, smooth regions, and global context.

This grouping operation is performed on learned latent hyperspectral features and characterizes complementary spatial-frequency response patterns rather than predefined physical wavelength intervals. The four components preserve the information of the projected feature and can reconstruct it through element-wise summation. For each spectral group X_g, we perform local spectral modeling. Instead of directly applying attention to the full feature map, we first partition it into N nonoverlapping spatial patches $M_g^{(j)}$, where P denotes the spatial size of each patch

$$X_g \to \left\{M_g^{(j)}\right\}_{j=1}^{N}, \quad M_g^{(j)} \in \mathbb{R}^{C_g\times P^2}. \tag{9}$$

This operation restricts spectral modeling within local spatial regions, which helps preserve spatial consistency and reduces computational complexity. To further remove spatial redundancy, GAP is applied along the spatial dimension

$$z_g^{(j)} = AvgPool\left(M_g^{(j)}\right) \in \mathbb{R}^{1\times C_g}. \tag{10}$$

The resulting vector $z_g^{(j)}$ serves as a compact descriptor of the spectral response within each local region. This step stabilizes spectral statistics and provides a robust basis for subsequent attention computation. However, purely local spectral modeling may lead to overly dispersed attention under complex degradations. To guide the model toward physically meaningful spectral patterns, we introduce a learnable spectral prompt $P_s \in \mathbb{R}^{L\times D}$, which acts as a set of unified low-rank spectral bases, where L denotes the number of bases and D represents their dimension. Instead of directly generating the query from $z_g^{(j)}$, we construct a prompt-guided query $Q_g^{(j)}$ by dynamically weighting these spectral bases, while the key $K_g^{(j)}$ and value $V_g^{(j)}$ are obtained via independent linear projections from $z_g^{(j)}$. The unified projection process is formulated as

$$\alpha_g^{(j)} = Soft\max\left(z_g^{(j)} W_\alpha\right) \tag{11}$$

$$Q_g^{(j)} = \alpha_g^{(j)} P_S W_q \tag{12}$$

$$K_g^{(j)} = z_g^{(j)} W_k, \quad V_g^{(j)} = z_g^{(j)} W_v \tag{13}$$

where $W_\alpha \in \mathbb{R}^{C_g\times L}$ is a learnable routing matrix that computes the contribution coefficient $\alpha_g^{(j)}$ for each spectral basis. Meanwhile, W_q, W_k, and W_v are learnable weight matrices that map the respective representations into a shared spectral attention subspace of dimension d. This design forces the attention mechanism to operate within a structured, prompt-guided subspace, effectively suppressing noise and improving generalization.

The spectral attention $A_g^{(j)}$ is then computed with a scaling factor d to produce the refined spectral descriptor $\hat{z}_g^{(j)} \in \mathbb{R}^{1\times C_g}$

$$A_g^{(j)} = Soft\max\left(\frac{Q_g^{(j)}\left(K_g^{(j)}\right)^\top}{\sqrt{d}}\right) \tag{14}$$

$$\hat{z}_g^{(j)} = A_g^{(j)} V_g^{(j)}. \tag{15}$$

This variable captures the modulated spectral characteristics of the jth local patch, where informative spectral dependencies are adaptively enhanced and degradation-related noise is suppressed under the guidance of the spectral prompt. The refined spectral representation is projected back to the original dimension using a learnable linear mapping W_p

$$\hat{M}_g^{(j)} = W_p\left(\hat{z}_g^{(j)}\right). \tag{16}$$

This vector $\hat{M}_g^{(j)}$ is used to reweight the original patch $M_g^{(j)}$, enabling adaptive enhancement or suppression of spectral responses. All patches are then rearranged to reconstruct the enhanced feature $\hat{X}_g$. After intragroup modeling, the enhanced features $\{\hat{X}_g\}_{g=1}^{G}$ are adaptively modulated according to their importance. Since different degradations affect spectral components differently, a fixed group contribution is insufficient. To address this, we design the semantic prompt P_t to dynamically determine the importance of each spectral feature group via a multilayer perceptron (MLP)

$$w_g = Softmax\left(\mathrm{MLP}\left(P_t\right)\right) \tag{17}$$

where $g \in S$, $S = \{high, mid, low, global\}$ denotes the four spectral feature groups. Each enhanced group feature is then modulated by its corresponding weight

$$\tilde{X}_g = w_g \hat{X}_g. \tag{18}$$

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The final output is obtained by concatenating all modulated group features

$$X_{\text{out}} = \text{Concat}\left(\tilde{X}_1, \tilde{X}_2, \ldots, \tilde{X}_G\right). \tag{19}$$

This design enables degradation-aware spectral modulation. For example, under heavy noise, the model can automatically reduce the contribution of high-frequency groups while relying more on stable low-frequency information. In summary, FPSM integrates frequency-aware grouping, prompt-guided spectral attention, and adaptive fusion into a unified framework. It allows the network to explicitly model spectral diversity and dynamically adjust feature contributions, leading to improved robustness under complex degradations.

E. Semantic–Visual Dynamic Prompting

To enable adaptive enhancement under complex and diverse degradation conditions, we propose a cross-modal SVDP mechanism, as shown in Fig. 4(c). SVDP integrates input-specific degradation evidence, structured semantic priors, and learnable visual patterns. Specifically, DPN estimates a soft degradation distribution from the input HSI. This distribution adaptively aggregates CLIP-based semantic anchors into a continuous semantic prompt, which further retrieves relevant visual patterns for restoration.

Given a degraded HSI, DPN first estimates a degradation-aware prior. It adopts a ResNet-34 [36] backbone augmented with fast Fourier convolution (FFC) operators to jointly capture spatial structures and frequency-domain characteristics. Instead of producing a hard degradation label, DPN predicts a soft probability distribution over predefined degradation categories

$$p = \left[p_1, p_2, \ldots, p_k\right] \tag{20}$$

where k denotes the number of predefined degradation categories and p_i is the confidence score of the ith category. Different confidence combinations allow p to describe ambiguous, intermediate, and mixed degradation conditions.

The probability vector provides input-conditioned degradation evidence, but it is not sufficiently expressive for direct feature modulation. Directly using DPN features may introduce scene-dependent spatial content into the prompt and make degradation-related guidance difficult to isolate. Independent learnable degradation embeddings can provide category-specific representations, but they treat different degradation categories as separate parameters and do not explicitly preserve their semantic relationships. We therefore introduce CLIP-based textual anchors to organize the DPN prediction in a structured degradation-semantic space.

For each degradation category, a textual description T_i is encoded by a frozen CLIP [33] text encoder $f_{\text{CLIP}}(\cdot)$. The resulting semantic anchors are weighted by the probability distribution predicted by DPN

$$T_{\text{agg}} = \sum_{i=1}^{k} p_i f_{\text{CLIP}}\left(T_i\right). \tag{21}$$

CLIP is not utilized to predict the degradation category again. Instead, it provides a shared semantic space for interpreting the image-conditioned degradation evidence. Unlike independent learnable embeddings, the CLIP text embeddings organize different degradation descriptions within a semantically structured representation space. Their probability-weighted aggregation therefore preserves the relationships among degradation types and forms a continuous semantic condition. Since the aggregation weights are inferred from the input HSI, T_{agg} is not restricted to a fixed one-hot category and can represent intermediate states and mixed degradation patterns. The aggregated representation is then projected into the hyperspectral feature space

$$P_t = \phi_t\left(T_{\text{agg}}\right) \tag{22}$$

where $\phi_t(\cdot)$ denotes a learnable adaptation layer. $P_t \in \mathbb{R}^{N\times D}$ is the semantic prompt, with prompt length N and embedding dimension D.

Although P_t provides high-level degradation guidance, image enhancement also requires spatial–spectral patterns associated with the inferred degradation condition. We therefore maintain a learnable visual pattern library $M \in \mathbb{R}^{L\times D}$, which stores latent degradation-related visual prototypes. These prototypes provide restoration-oriented visual priors that complement the semantic representation.

To connect degradation semantics with visual restoration patterns, we introduce semantic-guided cross-attention. The semantic prompt P_t serves as the query, while the visual pattern library M provides the key and value

$$Q = P_t W_q, \quad K = MW_k, \quad V = MW_v \tag{23}$$

$$P_v = Soft\max\left(\frac{QK^{\top}}{\sqrt{d}}\right)V \tag{24}$$

where W_q, W_k, and W_v are learnable projection matrices, and d denotes the scaling dimension. In this process, P_t specifies the degradation condition to be addressed, whereas M provides trainable spatial–spectral patterns associated with restoration. The cross-attention operation therefore converts high-level degradation semantics into visual guidance that can directly participate in feature reconstruction. The resulting visual prompt P_v can be interpreted as a semantic-conditioned retrieval result from the visual pattern library.

Finally, P_v is injected into the decoder through skip connections. At the nth stage, the encoder feature F_n^e is concatenated with the visual prompt and projected for feature fusion

$$\tilde{F}_n = f_{\text{proj}}\left(\text{Concat}\left(F_n^e, P_v\right)\right) \tag{25}$$

where $f_{\text{proj}}(\cdot)$ denotes a learnable projection that aggregates the concatenated representations. The prompt-augmented feature $\tilde{F}_n$ is then fed into the corresponding decoder block.

Through this design, DPN captures sample-specific degradation evidence, CLIP organizes it into a continuous and semantically structured representation, and the semantic prompt retrieves restoration-relevant visual patterns. This semantic–visual interaction enables the decoder to adapt its feature refinement to each input while preserving the spatial–spectral structures required for accurate HSI enhancement.

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IV. Experiments

A. Experimental Setup

1) Dataset and Data Preprocessing: To verify the generalization performance of the model under different imaging environments and ground characteristics, we construct a comprehensive hyperspectral dataset system covering natural scenes and remote sensing scenes. Since these two types of HSIs are acquired by different imaging systems and have different spectral characteristics, they are processed under two spectral-domain benchmark settings.

a) Natural HSI datasets: Experiments are conducted on two widely used natural-scene benchmarks, ARAD [37] and ICVL [6]. These datasets are acquired in the visible range and are represented with 31 spectral bands. Following standard protocols, ARAD is split into 900/50/50 images for training, validation, and testing, respectively. ICVL contains 201 images, with 100 used for training and 50 for testing. The training and testing sets are spatially disjoint. All training and testing patches are cropped to $64 \times 64 \times 31$. No spectral resampling is applied to the natural-scene HSI setting.

b) Remote sensing HSI datasets: We apply a diverse collection of remote sensing datasets, including WDC [28], PaviaC [28], XIONGAN, PaviaU [38], Chikusei, Houston, Berlin, Urban, and EO-1 Hyperion [31]. These datasets are acquired by different airborne or satellite sensors and exhibit substantial variations in band number, wavelength range, and spectral response. Directly training a unified remote sensing model on datasets with inconsistent spectral dimensions is impractical. Therefore, before degradation synthesis, training, and evaluation, remote sensing HSIs are standardized to a unified 100-band spectral grid over 400–1000 nm, following common practice in hyperspectral data preprocessing [31]. This resampling is a dataset-level standardization step, rather than a method-specific configuration or part of the restoration operation. In each experiment, the degraded input and clean target are generated on the same spectral grid, and the restored output preserves the same number of bands as the input.

2) Data Partitioning and Augmentation Strategies: For remote sensing datasets, we adopt a spatial-region-based split, where approximately 80% of the area is used for training and the remaining for testing. For large-scale scenes, a fixed or randomly selected central region is reserved as the test set, and the remaining regions are used for training.

To enhance the robustness of the model to multiscale spatial structures, we employ multiscale training. Specifically, training samples are constructed from three spatial scales, including the original resolution, and images downsampled by factors of 0.5 and 0.25. A sliding-window cropping scheme is then applied to extract training patches of size 64×64. All samples are normalized using min–max normalization. The processed data are subsequently packaged into the LMDB [39] database format. This storage strategy reduces I/O overhead during training and improves data loading throughput.

3) Degradation Simulation and Real-World Evaluation: During training, we adopt an on-the-fly degradation strategy. Each input sample is randomly corrupted using a combination of degradations, including noise, blur, downsampling, masking, and band corruption. In addition, data augmentation techniques such as random flipping and rotation are applied to further improve the robustness of the model to geometric transformations.

To validate the effectiveness of the proposed framework in practical scenarios, experiments are conducted on the Urban [31] dataset for real-noise denoising and the EO-1 Hyperion dataset for real-world dehazing. In these real-scene evaluations, the model performs inference on real degraded observations without introducing any synthetic degradations. For consistency with the remote sensing benchmark setting, low-quality spectral bands are first removed, and the valid spectral information is mapped to the corresponding evaluation spectral grid. This preprocessing is used to ensure compatible spectral dimensions and reliable evaluation, and does not synthesize additional degradation.

4) Evaluation Metrics: We apply peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) as the main reference-based metrics for evaluating pixel-wise reconstruction fidelity and structural similarity. Both metrics are computed per spectral band and then averaged across bands and test samples. To further evaluate spectral fidelity, we additionally report spectral angle mapper (SAM), root mean square error (RMSE), and mean relative absolute error (MRAE) in the spectral-fidelity analysis. SAM measures angular differences between reconstructed and reference spectra, while RMSE and MRAE quantify absolute and relative spectral reconstruction errors. For real-world HSI scenarios without clean reference images, reference-based metrics cannot be directly computed. Therefore, we further adopt no-reference HSI quality metrics, including QSFL and R-NHSIQA, and conduct physical consistency analyses to assess real-world restoration quality.

5) Baseline: To comprehensively evaluate the effectiveness of the proposed method across various HSI enhancement tasks, we compare the proposed method with representative deep learning methods, including both task-specific enhancement models and all-in-one enhancement frameworks.

For task-specific comparisons, we select representative networks optimized for individual inverse problems, such as SERT [40] for noise removal and MB [41] for dehaze. In addition, we compare our approach with several representative all-in-one HSI enhancement methods, including PromptHSI [32], InstructIR [23], HAIR [19], OneRestore [22], and MP-HSIR [31]. These methods aim to model multiple types of degradation within a unified framework, which makes them highly relevant for comparison with the proposed method. For fair comparison, all baseline methods and the proposed method are trained and evaluated using the same preprocessed hyperspectral cubes, spectral grids, degradation synthesis pipeline, train/test splits, and metric computation protocol. Within each spectral domain, a single unified model with one parameter set is trained to handle multiple degradation types: the natural-scene model operates on 31-band inputs, whereas the remote sensing model uses standardized 100-band inputs. The two models share the same architecture and proposed modules, with different training schedules applied to accommodate differences in spectral dimensionality, data scale, and restoration difficulty.

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TABLE I

QUANTITATIVE COMPARISON ON GAUSSIAN DENOISING AND COMPLEX DENOISING TASKS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN PURPLE AND ORANGE, RESPECTIVELY

Type	Method	*Gaussian Denoising ($\sigma = 70$)*			Method	*Complex Denoising (Case=1, 2, 3, 4)*		
		ICVL [6]	ARAD [37]	XIONGAN [38]		ICVL [6]	ARAD [37]	WDC [28]
		PSNR / SSIM	PSNR / SSIM	PSNR / SSIM		PSNR / SSIM	PSNR / SSIM	PSNR / SSIM
Task Specific	SERT [40]	34.55 / 0.9040	34.09 / 0.9050	31.44 / 0.7582	SERT [40]	37.88 / 0.8601	34.05 / 0.8402	24.16 / 0.6325
	HIRDiff [29]	38.12 / 0.9305	37.04 / 0.9210	27.01 / 0.7201	HIRDiff [29]	42.01 / 0.9702	42.15 / 0.9715	28.55 / 0.8102
	Volformer [27]	39.48 / 0.9492	36.54 / 0.9326	26.88 / 0.7150	Volformer [27]	41.55 / 0.9680	41.70 / 0.9690	28.80 / 0.8150
	LDERT [6]	39.83 / 0.9509	39.28 / 0.9457	35.14 / 0.8504	LDERT [6]	42.82 / 0.9767	42.27 / 0.9709	29.41 / 0.8148
All-in-One	PromptHSI [32]	33.39 / 0.8083	31.45 / 0.7599	27.66 / 0.6578	PromptHSI [32]	35.64 / 0.8855	33.31 / 0.8391	30.83 / 0.8501
	InstructIR [23]	37.55 / 0.9271	35.13 / 0.9013	24.18 / 0.6459	InstructIR [23]	39.28 / 0.9476	36.71 / 0.9270	27.49 / 0.7872
	HAIR [19]	33.46 / 0.8360	32.90 / 0.8210	25.32 / 0.6490	HAIR [19]	34.50 / 0.8710	33.21 / 0.8350	29.44 / 0.8230
	OneRestore [22]	32.26 / 0.8150	30.50 / 0.7400	24.11 / 0.6200	OneRestore [22]	33.74 / 0.8500	31.58 / 0.8100	27.52 / 0.8000
	MP-HSIR [31]	39.56 / 0.9500	37.36 / 0.9364	31.23 / 0.8014	MP-HSIR [31]	42.49 / 0.9718	40.67 / 0.9676	32.39 / 0.8850
	Ours	41.12 / 0.9650	39.47 / 0.9600	37.17 / 0.8745	Ours	43.04 / 0.9770	43.42 / 0.9770	34.99 / 0.9305

B. Implementation Details

All experiments are implemented in PyTorch and conducted on an NVIDIA RTX 5090 GPU. The model is optimized using the AdamW optimizer with parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$. We adopt an ℓ_1 loss for supervised training, and the batch size is set to 16. To account for the distribution discrepancy between natural scenes and remote sensing scenarios, we adopt differentiated training strategies as follows.

1) *Natural Scenes:* The model is trained for 100 epochs. The initial learning rate is set to 2×10^{-4} and is gradually decayed to 1×10^{-6} using a cosine annealing schedule.
2) *Remote Sensing:* Considering the large data scale and high spectral dimensionality, the training duration is extended to 300 epochs. The initial learning rate is set to 1×10^{-4}. In addition, to better capture the complex spectral correlations in remote sensing data, the channel width of the unified model is expanded by a factor of 1.5. For all baseline methods, hyperparameters are initialized according to their official configurations when available, and all methods are fine-tuned under the same benchmark protocol.

C. Comparison With State-of-the-Art Methods

1) Quantitative Comparison: We compare the proposed method with SOTA HSI enhancement methods across multiple degradation tasks. Table I reports quantitative results against six unified models and 22 task-specific methods over seven enhancement tasks.

In Gaussian deblurring and super-resolution tasks, as shown in Table II, the proposed method consistently achieves the best performance in both PSNR and SSIM. In the image inpainting task, as shown in Table III, the proposed method achieves 50.07-dB PSNR on the ICVL dataset, outperforming PromptHSI by 16.29 dB. On the Chikusei dataset, it reaches 42.08 dB, exceeding the second-best method by 5.39 dB. These results demonstrate its strong adaptability across diverse degradation scenarios. Notably, under high noise levels, the model maintains stable spectral reconstruction quality. This indicates robust spectral reconstruction capability. The improvement mainly comes from effective spectral modeling and degradation-aware prompt guidance.

2) Qualitative Comparison: To provide a clearer evaluation of enhancement quality, Fig. 5 presents visual comparisons on dehazing, image inpainting, and complex noise removal tasks. The differences are highlighted using zoomed-in view regions (red boxes), detail-enhanced views (blue boxes), and residual maps (yellow boxes).

In the dehazing task, MB [41] and DCMPNet [4] fail to fully correct the spectral shift caused by atmospheric scattering. This results in color distortion and blurred distant objects. The proposed method, guided by text-visual prompts, models haze degradation more accurately. It restores realistic colors and preserves fine details in distant regions. In the inpainting task, PromptHSI [32] and InstructIR [23] exhibit structural distortions and texture discontinuities under a high masking ratio (0.9). The proposed method produces filled regions that are consistent with surrounding areas in both texture and spectral continuity and restores fine details more naturally. In the complex noise removal task, SERT [40] and InstructIR [23] remove part of the noise. However, residual high-frequency perturbations lead to blurred textures. In contrast, the proposed method removes noise more thoroughly while preserving sharp object boundaries. The residual maps show minimal structural errors. Across all tasks, the Residual maps further confirm that our method produces lower and more uniform reconstruction errors. These observations are consistent with the quantitative results.

3) Real Scenario Comparison: To evaluate the robustness and generalization ability of the proposed framework beyond predefined synthetic degradations, we conduct experiments on two real hyperspectral datasets, Urban [31] and EO-1 Hyperion, together with controlled evaluations covering degradation types, severity levels, and combinations beyond the training configurations. The real observations contain acquisition-related degradations with varying severities and complex combinations, while the controlled experiments

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TABLE II

QUANTITATIVE COMPARISON ON GAUSSIAN DEBLURRING AND SUPER-RESOLUTION TASKS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN PURPLE AND ORANGE, RESPECTIVELY

Type	**Method**	*Gaussian Deblurring (radius=21/15)*			**Method**	*Super resolution (scale=8, 4, 2)*		
		ICVL [6]	Paviac [28]	Eagle [31]		ARAD [37]	Paviau [38]	Houston [31]
		PSNR / SSIM	PSNR / SSIM	PSNR / SSIM		PSNR / SSIM	PSNR / SSIM	PSNR / SSIM
Task Specific	Efficient [42]	46.03 / 0.9880	37.13 / 0.9110	39.86 / 0.9250	SRFormer [43]	31.02 / 0.7510	27.80 / 0.6041	28.80 / 0.6246
	MLWNet [44]	46.67 / 0.9897	36.01 / 0.8790	40.14 / 0.9280	SNLSR [45]	31.05 / 0.7980	26.45 / 0.5084	26.99 / 0.6023
	LoFormer [46]	47.15 / 0.9890	37.72 / 0.9170	42.59 / 0.9620	MAN [47]	29.88 / 0.7110	26.92 / 0.5995	28.03 / 0.6285
	BlurDM [48]	35.52 / 0.8924	35.88 / 0.8896	39.92 / 0.9456	FCA2 [49]	30.64 / 0.8401	27.95 / 0.6056	29.70 / 0.6800
All-in-One	PromptHSI [32]	38.69 / 0.9700	36.41 / 0.8952	40.81 / 0.9496	PromptHSI [32]	29.61 / 0.7968	30.49 / 0.7240	31.75 / 0.7777
	InstructIR [23]	39.16 / 0.9613	33.60 / 0.8290	37.64 / 0.9111	InstructIR [23]	30.10 / 0.8280	29.72 / 0.6943	30.79 / 0.7402
	HAIR [19]	46.45 / 0.9880	36.45 / 0.8998	40.85 / 0.9514	HAIR [19]	30.84 / 0.7300	27.47 / 0.6230	28.51 / 0.6410
	OneRestore [22]	43.20 / 0.9150	34.67 / 0.8520	39.98 / 0.9510	OneRestore [22]	29.59 / 0.7896	25.66 / 0.7992	27.03 / 0.5849
	MP-HSIR [31]	47.60 / 0.9915	37.19 / 0.9047	41.36 / 0.9519	MP-HSIR [31]	31.34 / 0.8428	28.38 / 0.6296	29.92 / 0.6000
	Ours	49.00 / 0.9931	37.93 / 0.9190	42.77 / 0.9640	Ours	38.22 / 0.9593	30.96 / 0.7375	32.06 / 0.7900

TABLE III

QUANTITATIVE COMPARISON ON INPAINTING, DEHAZING, AND COMPLETION TASKS. BLANK ENTRIES ARE DENOTED BY "–." THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN PURPLE AND ORANGE, RESPECTIVELY

Method	*Inpainting (mask=0.9)*		**Method**	*Dehazing (omega=1)*		**Method**	*Completion (rate=0.3)*	
	ICVL [6]	Chikusei [31]		Paviau [38]	Eagle [31]		ARAD [37]	Berlin [31]
	PSNR / SSIM	PSNR / SSIM		PSNR / SSIM	PSNR / SSIM		PSNR / SSIM	PSNR / SSIM
DDS2M [28]	42.18 / 0.9690	34.94 / 0.8870	SCANet [50]	32.34 / 0.9262	37.18 / 0.9746	ManbaIR [51]	49.26 / 0.9950	35.10 / 0.8862
HIRDiff [29]	42.19 / 0.9830	36.40 / 0.9322	DCMPNet [4]	32.44 / 0.9433	36.77 / 0.9737	-	-	-
-	-	-	MB [41]	31.99 / 0.7830	35.03 / 0.9410	-	-	-
PromptHSI [32]	33.78 / 0.8966	35.51 / 0.9103	PromptHSI [32]	32.43 / 0.9568	34.14 / 0.9702	PromptHSI [32]	35.52 / 0.9439	32.68 / 0.8174
InstructIR [23]	38.71 / 0.9609	27.59 / 0.6129	InstructIR [23]	32.73 / 0.7920	37.64 / 0.9111	InstructIR [23]	37.93 / 0.9656	23.76 / 0.7125
HAIR [19]	43.02 / 0.9610	31.06 / 0.7660	HAIR [19]	30.55 / 0.6940	35.86 / 0.9620	HAIR [19]	44.08 / 0.9820	24.22 / 0.7730
OneRestore [22]	41.48 / 0.9550	30.33 / 0.7590	OneRestore [22]	30.26 / 0.6880	33.65 / 0.9470	OneRestore [22]	47.34 / 0.9940	21.97 / 0.6930
MP-HSIR [31]	48.35 / 0.9952	36.69 / 0.9309	MP-HSIR [31]	29.15 / 0.9726	35.51 / 0.9897	MP-HSIR [31]	55.15 / 0.9985	32.56 / 0.9081
Ours	50.07 / 0.9947	42.08 / 0.9753	Ours	33.67 / 0.9590	38.66 / 0.9870	Ours	55.71 / 0.9952	35.98 / 0.9300

isolate different forms of degradation-distribution shift. These complementary settings assess adaptability to both practical acquisition conditions and diverse degradation configurations.

The EO-1 Hyperion scene was acquired by the Hyperion sensor onboard the NASA EO-1 satellite. After removing water-absorption and severely corrupted bands, 145 valid spectral bands are retained. The Urban dataset was acquired by the HYDICE sensor with a spatial size of 307×307, and 162 valid bands are used after standard preprocessing. In both real-scene evaluations, all models are directly applied to the degraded observations without additional degradation synthesis, network fine-tuning, or explicit degradation labels.

The visual comparisons are shown in Fig. 5. On Urban, the input HSI contains mixed noise and dense stripe artifacts. Several competing methods retain visible stripes or over-smooth roads and building boundaries, while PromptHSI [32] introduces noticeable pseudo-color shifts. In contrast, the proposed method more effectively suppresses structured stripes while preserving clearer spatial textures and more stable pseudo-color representations. On the EO-1 Hyperion Botswana scene, spatially varying thin haze reduces global contrast and obscures structural details. Existing methods either leave residual haze, reduce color saturation, or introduce local color inconsistency, whereas the proposed method restores clearer structures and higher contrast while maintaining more consistent spectral-color characteristics.

To assess restoration quality without clean References, we report two no-reference HSI quality metrics, QSFL [52] and R-NHSIQA [53], together with band-column stripe responses and pixel-wise spectral-fluctuation distributions for spatial–spectral consistency analysis.

As shown in Table IV, the proposed method achieves the lowest QSFL and R-NHSIQA values on both datasets, indicating improved no-reference restoration quality under real acquisition degradations. Fig. 6 presents the band-column stripe-response heatmaps on Urban. The degraded input and several competing methods exhibit dense high-response patterns or pronounced band-dependent fluctuations. In contrast, the proposed method produces lower and sparser responses

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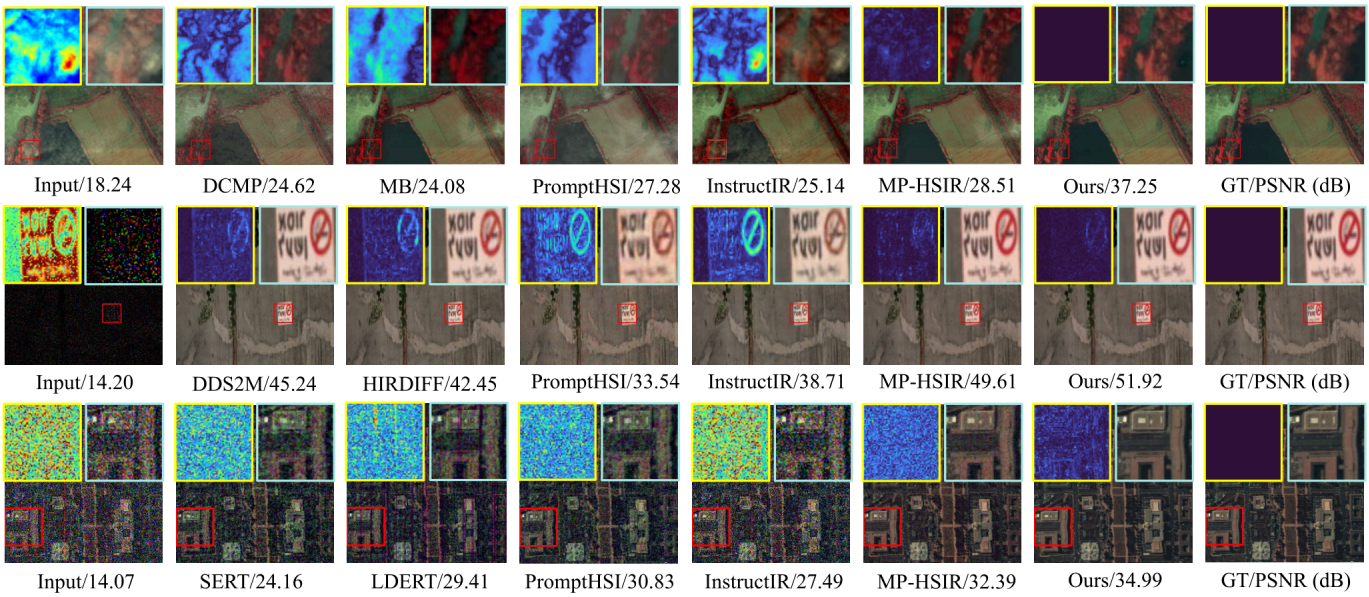

Fig. 5. **[All-in-one]** Visual comparison results on **dehazing**, **inpainting**, and **complex denoising**. The zoomed-in view regions are highlighted with red boxes, the corresponding zoomed-in view results are marked with blue boxes, and the residual maps are outlined with yellow boxes.

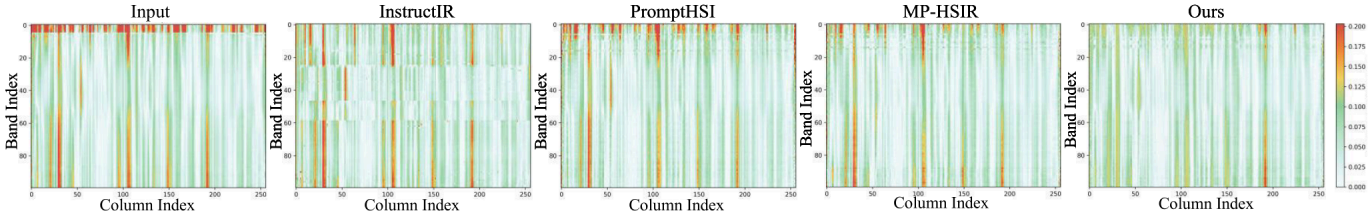

Fig. 6. No-reference band-column stripe-response heatmaps on the real Urban dataset. The horizontal and vertical axes denote the spatial column and spectral band indices, respectively. The color represents the normalized column-wise stripe fluctuation estimated from each restored HSI rather than a residual with respect to a clean reference. Warmer colors indicate stronger remaining stripe responses.

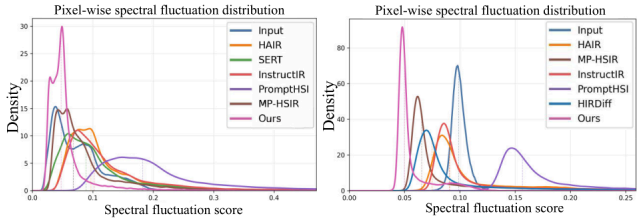

Fig. 7. Pixel-wise spectral-fluctuation distributions on the real-world Urban and EO-1 Hyperion datasets. The horizontal axis denotes the spectral-fluctuation score, and the vertical axis denotes the probability density. A distribution concentrated in the lower-fluctuation region indicates fewer abnormal spectral oscillations and improved physical spectral consistency.

TABLE IV

NO-REFERENCE QUALITY ASSESSMENT ON THE REAL-WORLD URBAN AND EO-1 HYPERION DATASETS. LOWER QSFL AND R-NHSIQA VALUES INDICATE BETTER NO-REFERENCE HSI QUALITY

Dataset	Method	QSFL [52] ↓	R-NHSIQA [53] ↓
Urban	SERT [40]	15.66	3.15
	HIRDiff [29]	15.21	2.78
	HAIR [19]	14.95	3.32
	InstructIR [23]	15.36	3.63
	PromptHSI [32]	12.34	2.24
	MP-HSIR [31]	11.42	1.83
	Ours	**10.98**	**1.69**
EO-1 Hyperion	DCMP [4]	20.11	3.71
	MB [41]	18.52	3.66
	HAIR [19]	17.99	2.81
	InstructIR [23]	19.71	3.43
	PromptHSI [32]	18.13	2.51
	MP-HSIR [31]	16.54	2.08
	Ours	**15.69**	**1.72**

across most bands and columns, demonstrating more effective suppression of structured stripe artifacts.

Fig. 7 shows the pixel-wise spectral-fluctuation distributions on Urban and EO-1 Hyperion. Competing methods tend to retain broader distributions or abnormal high-fluctuation tails, indicating unstable local spectral behavior. The proposed method yields a more compact distribution concentrated in the lower-fluctuation region, suggesting fewer abnormal spectral oscillations without excessively suppressing meaningful spatial–spectral variations.

As a controlled complement to the real-world evaluation, we further test all methods under three settings involving degradation types, severity levels, and combinations beyond the training configurations. This protocol isolates different forms of degradation-distribution shift that cannot be controlled separately in real observations. As shown in Fig. 8, competing methods may retain residual corruption, introduce local pseudo-color inconsistency, under-correct severe

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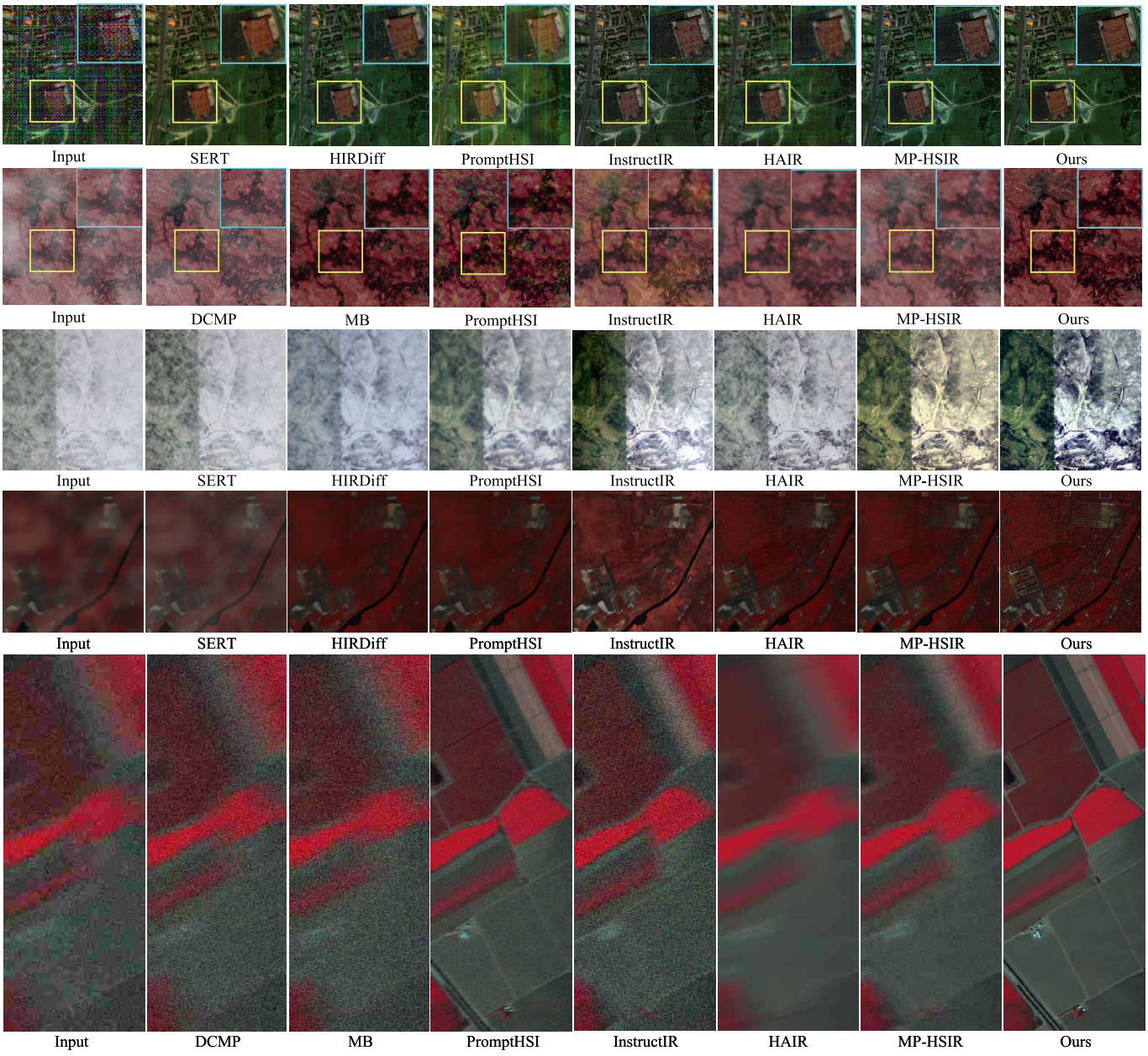

Fig. 8. **[Real scenario]** Visual comparisons under real-world and diverse degradation scenarios. The real-world groups include Urban with mixed noise and stripe artifacts and EO-1 Hyperion with spatially varying haze. All methods are evaluated using the trained weights directly, without fine-tuning, explicit degradation labels, or manual instructions. The proposed method produces clearer structures, fewer residual artifacts, and more stable pseudo-color representations across both real-world and diverse degradation settings.

degradations, or over-smooth fine spatial details. These limitations are more evident for complex degradation combinations, where spatial and spectral distortions are coupled. By contrast, the proposed method consistently reconstructs clearer structures, preserves more local details, and produces more stable pseudo-color results.

Overall, the real-scene experiments demonstrate the robustness of the proposed framework to practical acquisition degradations across different sensors, while the controlled evaluations provide further evidence of its adaptability to degradation types, severity levels, and combinations beyond the training configurations. These results indicate that the framework is not restricted to fixed one-to-one mappings between predefined degradation categories and restoration operations, but can adjust its restoration behavior under mixed, diverse, and challenging degradation conditions. We emphasize that the observed generalization is confined to the considered HSI degradation space rather than arbitrary open-world degradations.

4) Spectral Fidelity: Fig. 9 compares the spectral reflectance curves of different methods across representative enhancement tasks. The proposed method shows closer alignment with the ground truth spectra and produces lower spectral deviations across most bands. In complex noise removal, the reconstructed spectra remain smooth and stable, while PromptHSI and InstructIR show noticeable deviations in the

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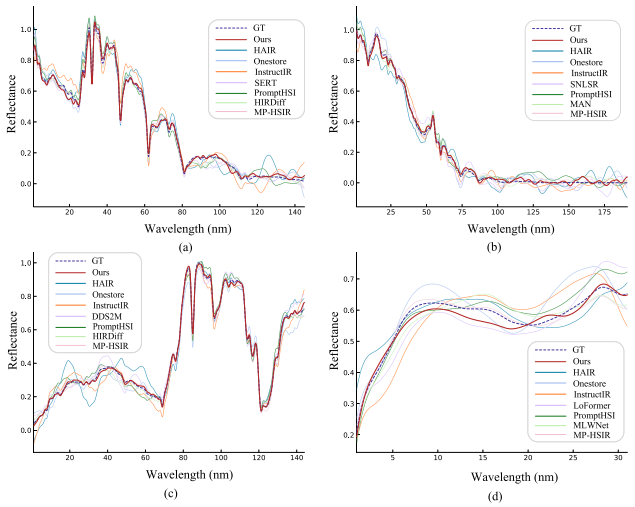

Fig. 9. Comparison of spectral reflectance curves reconstructed by different methods across representative enhancement tasks. (a) Denoising on the Botswana dataset. (b) Super-resolution on the WDC dataset. (c) Inpainting on the Houston dataset. (d) Deblurring on the ARAD dataset. The proposed method shows better spectral fidelity by reducing spectral distortion and preserving material-related spectral characteristics.

TABLE V

QUANTITATIVE COMPARISON OF SPECTRAL FIDELITY METRICS UNDER FOUR REPRESENTATIVE ENHANCEMENT TASKS. LOWER SAM, RMSE, AND MRAE VALUES INDICATE BETTER SPECTRAL FIDELITY, AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

WDC–Gaussian Denoising				WDC–Super-resolution			
Method	**SAM ↓**	**RMSE ↓**	**MRAE ↓**	**Method**	**SAM ↓**	**RMSE ↓**	**MRAE ↓**
SERT [40]	16.769	0.103	1.104	MAN [47]	10.564	0.055	0.377
LDERT [6]	13.507	0.0685	0.957	SNLSR [45]	8.009	**0.052**	**0.230**
PromptHSI [32]	17.450	0.061	0.548	PromptHSI [32]	9.549	0.093	0.805
MP-HSIR [31]	14.953	0.054	**0.161**	MP-HSIR [31]	9.353	0.092	0.619
Ours	**13.027**	**0.047**	0.547	**Ours**	**6.348**	0.067	0.355
Houston–Inpainting				**ARAD–Gaussian Deblurring**			
Method	**SAM ↓**	**RMSE ↓**	**MRAE ↓**	**Method**	**SAM ↓**	**RMSE ↓**	**MRAE ↓**
DDS2M [28]	6.750	0.052	1.474	LoFormer [46]	4.842	0.039	0.278
HIRDiff [29]	5.965	0.051	0.914	MLWNet [44]	4.867	0.039	0.271
PromptHSI [32]	8.981	0.040	1.004	PromptHSI [32]	4.449	0.030	0.121
MP-HSIR [31]	8.259	0.036	0.748	MP-HSIR [31]	4.270	0.028	0.116
Ours	**5.276**	**0.021**	**0.357**	**Ours**	**3.614**	**0.025**	**0.106**

short-wavelength region. For image inpainting, our method better preserves characteristic spectral peaks, whereas competing methods tend to introduce spectral distortions. This advantage mainly benefits from the spectral prior embedded in the spectral prompt, which helps enforce spectral consistency during enhancement.

To provide quantitative evidence, we further evaluate spectral fidelity using three spectral-related metrics: SAM, RMSE, and MRAE. Unlike PSNR and SSIM, which mainly measure pixel-wise fidelity and structural similarity, these metrics directly reflect spectral angle preservation, spectral reconstruction error, and material signature consistency. Lower values indicate better spectral fidelity.

As reported in Table V, the quantitative evaluation is conducted on four representative tasks, including Gaussian denoising, super-resolution, inpainting, and Gaussian deblurring.

TABLE VI

DOWNSTREAM LAND-COVER CLASSIFICATION PERFORMANCE OF RESTORED HSIS. THE DEGRADED INPUT AND CLEAN GROUND TRUTH ARE INCLUDED AS LOWER-BOUND AND UPPER-BOUND REFERENCES, RESPECTIVELY

Type	Method	OA (%)	AA (%)	Kappa (%)
Input	Degraded	52.64	52.58	38.07
Task-specific	LDERT [6]	75.15	73.64	65.86
	LoFormer [46]	76.55	75.88	68.06
	MambaIR [51]	76.67	76.13	68.20
All-in-One	PromptHSI [32]	77.62	77.91	69.23
	MP-HSIR [31]	83.44	79.37	78.13
	Ours	**84.32**	**87.07**	**79.04**

Since the Botswana data used for curve visualization do not provide clean references for quantitative evaluation, Gaussian denoising metrics are computed on the WDC dataset instead. The proposed method achieves the best SAM values across all tested settings and obtains the best or competitive RMSE and MRAE results, especially for inpainting and deblurring where spatial corruption and spectral distortion are strongly coupled.

Overall, both the spectral curves and quantitative metrics demonstrate that the proposed method improves spectral angle preservation, material signature consistency, and spectral reconstruction accuracy. These results verify that the proposed spatial–spectral prompt-driven framework enhances not only visual quality and spatial reconstruction, but also the physical spectral fidelity of restored HSIs.

5) Downstream Land-Cover Classification Evaluation: To further evaluate the practical utility of restored HSIs, we conduct a downstream land-cover classification experiment. The restored images generated by different enhancement methods are used as inputs for land-cover classification. The classification performance is evaluated using overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. The degraded input is also reported as lower-bound references, respectively. All methods are evaluated under the same classification protocol for fair comparison.

As reported in Table VI, the degraded input results in poor classification performance because degradation disrupts both spatial structures and spectral signatures. After restoration, most methods improve the classification accuracy to different degrees. Compared with task-specific and all-in-one baselines, the proposed method achieves the best OA and Kappa values among the restored results, indicating more reliable overall land-cover recognition. This improvement suggests that the proposed framework better preserves discriminative spatial–spectral information, thereby providing more effective inputs for downstream classification and remote sensing interpretation.

The visual classification maps in Fig. 10 further support the quantitative comparison. The degraded input contains obvious category confusion and incomplete land-cover regions. Existing restoration methods can partially recover the classification structures, but they still produce fragmented regions, inaccurate boundaries, or local misclassifications. In contrast, the proposed method generates a classification map that is closer to the ground truth, with more complete land-cover regions,

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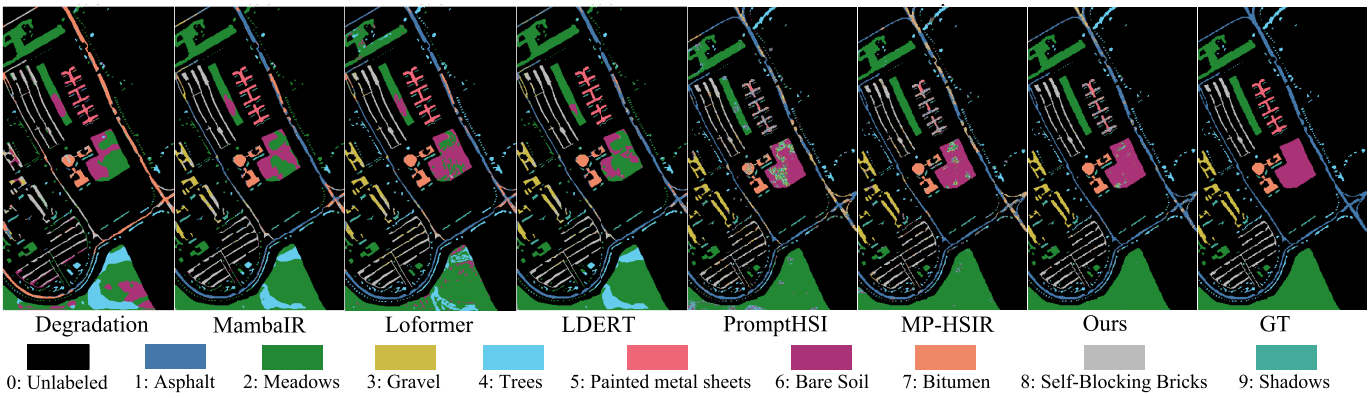

Fig. 10. Classification maps are obtained using degraded, restored, and clean HSIs. Compared with existing methods, the proposed method produces more complete land-cover regions, clearer structural boundaries, and fewer fragmented misclassifications, showing better consistency with the ground truth.

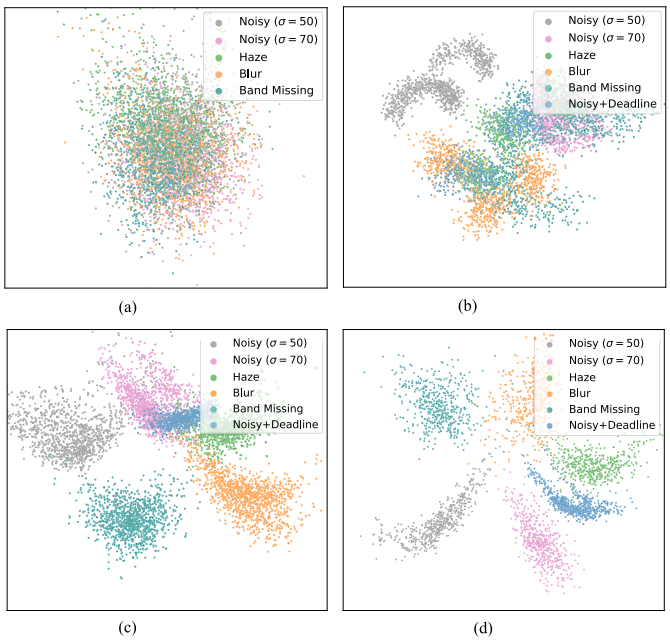

Fig. 11. t-SNE visualization of feature representations produced by different methods. Existing methods exhibit overlapping feature distributions, while the proposed method achieves more compact clusters and clearer separation. This indicates improved feature discriminability and better degradation-aware representation learning. (a) InstructIR. (b) PromptHSI. (c) MP-HSIR. (d) Ours.

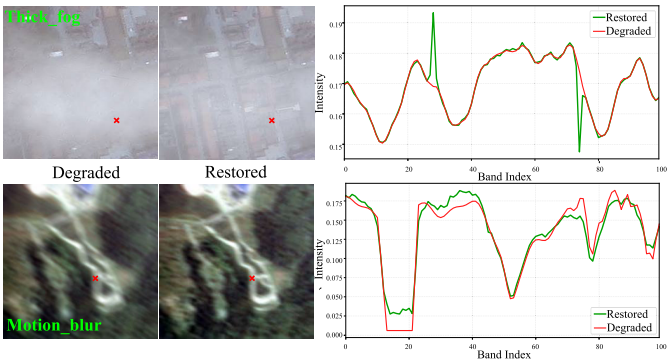

Fig. 12. Failure case analysis under extremely severe real-world degradations. The proposed method can recover major spatial structures under thick fog and motion blur, while residual spatial details and spectral deviations remain in heavily degraded regions, revealing the limitations of the current framework.

clearer object boundaries, and fewer scattered misclassifications.

6) Analysis of Feature Discriminability: To further evaluate the representation capability of different methods, we visualize the learned feature embeddings using t-SNE. The results are shown in Fig. 11. Existing methods produce feature distributions with noticeable overlap across different categories. This indicates limited discriminability and weak capability in capturing degradation-aware characteristics. The features tend to be scattered, and the boundaries between clusters are not well defined.

In contrast, the proposed method generates more compact feature clusters with clearer separation. The intraclass features are more consistent, while the interclass distance is significantly increased. This suggests that the model can better distinguish different underlying structures and degradation patterns. By integrating complementary prior information from multiple modalities, the model learns more informative and structured representations. As a result, the extracted features become more discriminative and robust. This observation is consistent with the quantitative results reported in Table III, where the proposed method achieves superior performance across multiple evaluation metrics.

D. Failure Case Analysis

Although the proposed method achieves promising restoration performance under various degradation conditions, extremely severe real-world degradations remain challenging. To further investigate the limitations of our framework, we present two representative failure cases under thick fog and motion blur in Fig. 12.

For thick fog and motion blur, the proposed method can alleviate the degradation effects and recover the main spatial structures. However, some details remain unrecovered in heavily degraded regions, and the restored spectral responses still exhibit deviations in several bands, indicating that extremely severe degradations may limit accurate spatial–spectral reconstruction.

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TABLE VII

COMPUTATIONAL COMPLEXITY COMPARISON ON THE REMOTE SENSING HSI ENHANCEMENT SETTING. PARAMS, FLOPS, AND INFERENCE TIME ARE REPORTED FOR REPRESENTATIVE METHODS AND ALL-IN-ONE ENHANCEMENT METHODS

Type	Method	Params (M)	FLOPs (G)	Inference Time (s)
Task-specific	SERT [40]	2.30	18.35	0.041
	HAIR [19]	17.28	6.46	0.068
All-in-One	InstructIR [23]	154.03	6.57	0.065
	PromptHSI [32]	50.89	21.91	0.282
	MP-HSIR [31]	30.91	32.74	0.083
	Ours	42.13	35.66	0.105

These failure cases reveal the limitation of the proposed framework under extremely severe degradation conditions. When degradations significantly damage spatial structures or distort spectral responses, the missing spatial–spectral information becomes difficult to recover solely from degraded observations. Future research will focus on developing stronger physical degradation modeling and more adaptive restoration strategies to further improve robustness in challenging real-world scenarios.

E. Computational Complexity Analysis

Since the proposed framework introduces multiple spatial, spectral, and prompt-based components, we further analyze its computational complexity. Specifically, we report the number of parameters, FLOPs, and inference time under the remote sensing HSI enhancement setting. The compared methods include representative task-specific restoration methods and recent all-in-one enhancement methods. All results are measured under the same input setting and hardware environment for fair comparison.

We further clarify the scope of the complexity calculation. The reported results for our method cover the complete online inference pipeline, including the restoration backbone, the degradation perception network, the online prompt-generation and fusion operations, and the visual prototype bank with its retrieval operation. As a learnable component of SVDP, the visual prototype bank is included in both the parameter and FLOPs calculations. The frozen CLIP text encoder, however, is used once before inference to encode the fixed degradation descriptions. These text embeddings are cached and reused for all test images. Therefore, the CLIP encoder is not executed or loaded as part of the per-image online restoration pipeline, and its parameters, FLOPs, and one-time encoding time are not included in Table VII.

As illustrated in Table VII, the proposed method contains 42.13-M parameters, requires 35.66-G FLOPs, and takes 0.105 s for inference. It uses substantially fewer parameters than InstructIR and is faster than PromptHSI, while introducing moderate additional cost compared with MP-HSIR and lightweight task-specific methods. This overhead arises from the joint modeling of degradation-aware, spatial, and spectral information, but remains controlled through rectangular spatial attention, grouped spectral modulation, and compact prototype-based prompt retrieval. Overall, the proposed method provides a reasonable balance between restoration capability and computational complexity.

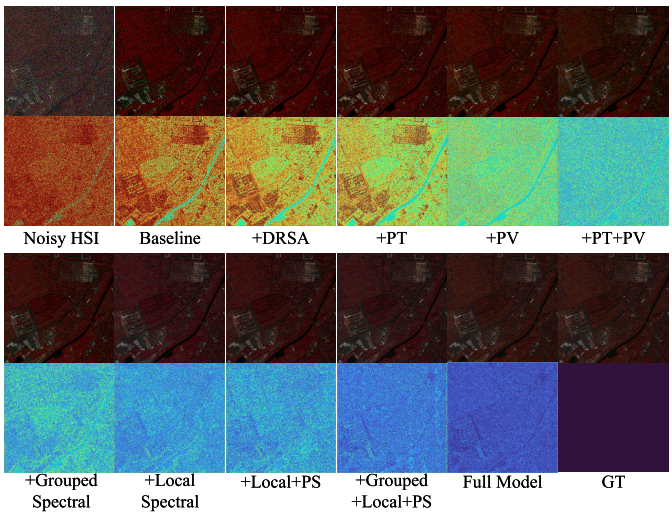

Fig. 13. Qualitative comparison of different components on the Berlin dataset.

TABLE VIII

ABLATION STUDY ON THE ARAD DATASET FOR GAUSSIAN DENOISING WITH NOISE LEVEL $\sigma = 70$. PSNR, SSIM, AND PARAMETER NUMBER ARE REPORTED TO EVALUATE THE CONTRIBUTION OF EACH COMPONENT

Stage	Method	PSNR	SSIM	Params (M)
I-0	Baseline	29.86	0.6671	14.6
I-1	+DRSA	31.31	0.7476	15.0
II-1	+Textual Prompt P_T	31.45	0.7519	15.0
II-2	+Visual Prompt P_V	33.92	0.8711	15.0
II-3	+P_T+P_V	34.04	0.8455	15.0
III-1	+Grouped Spectral+P_T+P_V	35.26	0.9096	17.7
III-2	+Local Spectral SA+P_T+P_V	35.69	0.9185	15.1
III-3	+Local Spectral SA+P_T+P_V+P_S	37.08	0.9339	15.3
Full Model		**37.23**	**0.9362**	**18.6**

F. Ablation Study

In this section, we conduct comprehensive ablation studies to validate the contribution of each component in the proposed prompt-driven spatial–spectral framework for HSI enhancement. Quantitative results are reported on the ARAD dataset using PSNR and SSIM as evaluation metrics. For better visual comparison, qualitative results are additionally provided on the Berlin dataset, which contains more complex spatial structures and degradation patterns.

We adopt a standard U-shaped architecture equipped with window-based self-attention as the baseline. Based on this baseline, different components are progressively introduced to analyze their individual contributions.

1) Effectiveness of DRSA: Standard square-window attention is limited in capturing anisotropic spatial structures. To address this issue, we replace it with the proposed DRSA. As shown in Table VIII, the introduction of DRSA yields a significant improvement of 1.45 dB in PSNR compared to the baseline. This gain demonstrates the effectiveness of directional rectangular attention in modeling long-range

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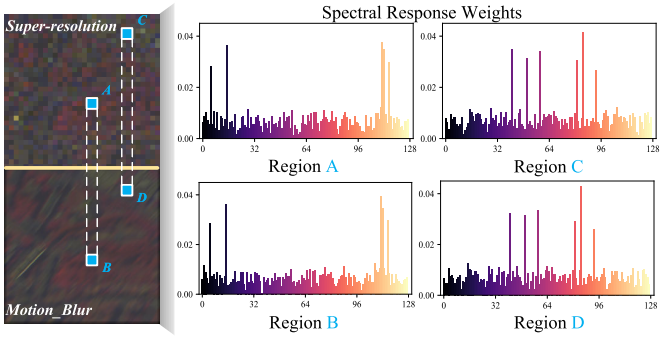

Fig. 14. Visualization of the task-specific activation for spectral patterns within the spectral prompt.

dependencies along dominant orientations. Furthermore, visual comparisons on the Berlin dataset in Fig. 13 further show that DRSA produces sharper edges and clearer structural details, while the baseline suffers from noticeable blurring artifacts. This confirms that DRSA enhances spatial structure recovery.

2) Effectiveness of SVDP: We then evaluate the contribution of the SVDP mechanism. Starting from the DRSA-based model, we first introduce textual prompts to construct Model II-1. This variant achieves a gain of 0.14 dB, indicating that high-level semantic descriptions provide useful global context for the enhancement process. Next, we replace textual prompts with visual prompts in Model II-2. The performance increases significantly, showing that visual prompts are more effective in capturing local degradation patterns. More importantly, Model II-3 achieves a significant improvement of 2.37 dB, demonstrating that the combination of textual and visual prompts outperforms relying on a single modality. This suggests that the two are complementary: textual prompts guide global semantic enhancement, while visual prompts refine local texture details. These results validate the effectiveness of the SVDP design, where multimodal priors are jointly encoded to address unified degradation scenarios.

3) Effectiveness of FPSM: We further analyze the proposed FPSM mechanism, which is designed to enhance spectral modeling. As shown in Stage III of Fig. 13, we analyze the hierarchical contributions of its key components, including global spectral grouping, local spectral self-attention, and spectral prompt selection. Built upon Model II-3, we first introduce global spectral grouping to construct Model III-1. This design reduces channel redundancy via grouping while improving the modeling of spectral correlations. As a comparison, replacing the global branch with local spectral self-attention (Model III-2) also leads to performance gains, owing to its ability to capture fine-grained interchannel dependencies within local spectral windows.

Furthermore, we incorporate the spectral prompt mechanism in Model III-3. Quantitative results demonstrate that the spectral prompt effectively retrieves the most relevant spectral atoms to modulate feature representations, leading to further improvements in spectral accuracy. The full model, integrating all components, achieves the best performance. We further visualize the activation distributions of the learnable spectral prototypes under super-resolution and motion blur degradations in Fig. 14. By comparing the same spatial regions across different degradation types, we observe that the activation patterns remain highly consistent. This indicates that the spectral prompts capture intrinsic spectral structures in HSIs. Meanwhile, FPSM effectively integrates global spectral consistency with local spectral continuity, enabling accurate reconstruction across all bands and maintaining robustness under diverse degradations.

TABLE IX

ABLATION STUDY ON THE EFFECT OF GROUPED NUMBER. THE OPTIMAL NUMBER OF GROUPS IS HIGHLIGHTED IN BOLD

Grouped Number	PSNR	SSIM	Params(M)
1	30.11	0.7038	15.9
2	36.69	0.9309	16.8
4	**42.08**	**0.9753**	**18.6**
8	41.86	0.9724	22.2

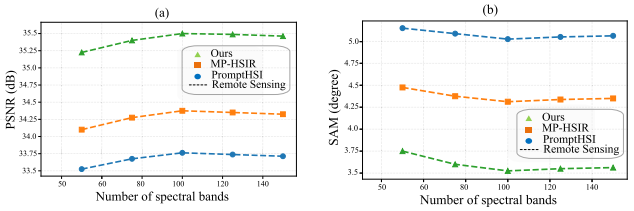

Fig. 15. Sensitivity analysis of spectral sampling configurations for remote sensing HSI enhancement. XIONGAN under the complex denoising setting is used as a representative case, with models separately trained and evaluated under each spectral-band configuration. The results demonstrate consistent (a) PSNR and (b) SAM performance across different spectral-band configurations within a reasonable range.

4) Effectiveness of the Group Number: To balance computational efficiency and representation capacity, we adopt a grouping strategy in the global spectral branch. To investigate the effect of the group number G, we train the model with different values of G. As reported in Table IX, the performance consistently improves as G increases and reaches the optimum at $G = 4$. This behavior suggests that an appropriate grouping scheme enables effective modeling of interchannel dependencies within subspaces while avoiding excessive parameter overhead. However, further increasing G to 8 leads to performance saturation or slight degradation. This indicates that overly fine-grained grouping may disrupt the continuity of global spectral information. Therefore, we set $G = 4$ as the default configuration in all experiments.

5) Spectral Sampling Sensitivity Analysis: To investigate the influence of spectral configurations on restoration performance, we conduct an additional spectral sampling sensitivity analysis. Specifically, models are separately trained and evaluated under each spectral-band configuration, while the wavelength range, degradation setting, data split, and overall training strategy are kept consistent. For this analysis, we conduct complex denoising experiments on XIONGAN [38], representing remote sensing HSI restoration scenarios.

Fig. 15 presents the results under different spectral band sampling settings. When the sampling rate is too low, the restoration performance declines. However, within

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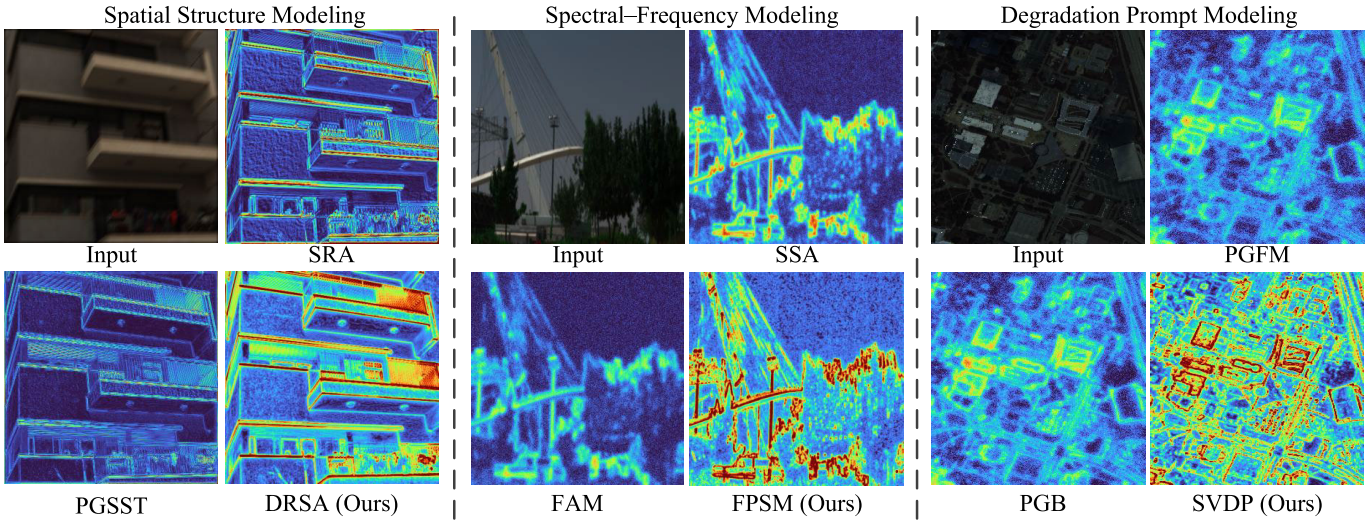

Fig. 16. Feature-response visualization for the component replacement study. Each map shows the response intensity of restoration-related features under a replacement setting. Warmer or brighter colors denote stronger feature responses, the proposed DRSA, FPSM, and SVDP produce stronger and more coherent responses on structural edges, spectral-frequency regions, and degradation-aware areas, respectively, indicating more effective feature activation than the alternative components.

TABLE X

PROMPT-SOURCE ABLATION ON THE HOUSTON DATASET. ALL VARIANTS KEEP THE SAME RESTORATION BACKBONE AND ONLY REPLACE THE PROMPT-GENERATION STRATEGY IN SVDP

Prompt Design	PSNR/SSIM/SAM	Params (M)
Learnable Embedding [21]	37.56/0.947/4.96	41.91
DPN Probability Prompt	38.02/0.954/4.86	42.03
DPN Feature Prompt	38.41/0.961/4.79	42.18
CLIP Semantic Prompt [54]	38.67/0.966/4.74	42.09
SVDP (Ours)	**39.30/0.973/4.69**	**42.13**

TABLE XI

COMPONENT REPLACEMENT STUDY ON THE HOUSTON DATASET. DR, FP, AND SV ARE ABBREVIATIONS FOR DRSA, FPSM, AND SVDP, RESPECTIVELY

Validation	Method	PSNR/SSIM/SAM	Params (M)
DRSA	SRA [40]+FP+SV	36.42/0.929/5.53	41.85
	PGSST [31]+FP+SV	36.16/0.926/5.64	42.77
FPSM	SSA [26]+DR+SV	34.54/0.944/5.84	41.53
	FAM [32]+DR+SV	33.51/0.854/6.04	42.69
SVDP	PGFM [32]+DR+FP	37.23/0.936/4.88	43.06
	PGB [21]+DR+FP	37.96/0.954/4.73	43.58
Ours	**DR+FP+SV**	**39.30/0.973/4.69**	**42.13**

a reasonable and controllable sampling range, all methods exhibit only marginal fluctuations in PSNR and SAM. This observation indicates that the restoration results are not strongly affected by a specific spectral sampling configuration. The selected spectral grids mainly serve as standardized representations for organizing heterogeneous HSI datasets, rather than providing a specific advantage for the restoration process. These results demonstrate that the effectiveness of our method is not dependent on a specific spectral configuration. Instead, the performance improvement mainly comes from the proposed spatial–spectral modeling and degradation-aware prompting mechanisms.

6) Necessity of CLIP-Based Semantic Prompting: To further verify the necessity of the proposed semantic–visual dynamic prompting strategy, we conduct a prompt-source ablation study on the Houston dataset, as shown in Table X. All variants share the same restoration backbone, DPN, degradation settings, and training protocol, and differ in the prompt-generation strategy. Learnable degradation embeddings [21] obtain the weakest performance due to the lack of input-dependent degradation perception and semantic priors. DPN probability and feature prompts improve the results by introducing degradation-aware cues, but their guidance remains relatively coarse. The CLIP semantic prompt further improves restoration performance by providing textual semantic anchors [54], yet it is still inferior to the full SVDP. By combining semantic anchors with visual degradation cues, SVDP achieves the best performance, demonstrating that semantic–visual prompt interaction provides more effective degradation-adaptive guidance than using learnable prompts or DPN-derived prompts alone.

7) Component Replacement Study: To further validate the necessity of the proposed components, we conduct a component replacement study on the Houston dataset. Different from standard ablation, this study replaces each proposed module with representative alternatives while keeping the remaining architecture unchanged. As reported in Table XI, all replacement variants lead to performance degradation, confirming that the proposed DRSA, FPSM, and SVDP are more suitable for unified HSI enhancement under complex degradations. Specifically, replacing DRSA weakens the recovery of directional structures and object boundaries. Replacing FPSM reduces spectral reconstruction accuracy, indicating the importance of prompt-guided spectral modulation. Replacing SVDP decreases degradation adaptability, showing that semantic–visual prompting provides more reliable restoration guidance. Fig. 16 further presents feature-response visualizations for different replacement settings. These maps show

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restoration-related feature activation. Thus, warmer or brighter regions indicate stronger feature responses. For spatial modeling, SRA [40] and PGSST [31] produce weaker or fragmented activations, while DRSA generates stronger and more continuous responses along edges and elongated structures. For spectral modeling, SSA [26] and FAM [32] show scattered activations, whereas FPSM produces more coherent responses in spectrally informative regions. For prompt modeling, PGFM [32] and PGB [21] either activate irrelevant areas or miss degradation-sensitive regions, while SVDP yields more consistent degradation-aware responses.

Overall, the component replacement results demonstrate that the proposed modules cannot be replaced by conventional alternatives without performance loss. The quantitative results and feature-response maps jointly show that DRSA, FPSM, and SVDP contribute complementary benefits to spatial structure recovery, spectral feature modulation, and degradation-aware guidance.

V. Conclusion

In this article, we proposed a prompt-driven spatial–spectral modeling framework for HSI enhancement via SVS multi-prompt learning. The proposed framework aims to achieve degradation-level unification within each spectral-domain setting and improve adaptability to diverse, mixed, and shifted degradation conditions within the considered HSI degradation space. By jointly integrating semantic–visual dynamic prompting, DRSA, and frequency-aware spectral modulation, the proposed method provides degradation-aware prior guidance, enhances anisotropic spatial structure recovery, and preserves spectral fidelity. Specifically, SVDP generates adaptive prompts by combining semantic anchors with visual degradation prototypes, DRSA improves the restoration of directional structures and fine boundaries, and FPSM strengthens spectral consistency by modeling frequency-dependent spectral responses. Extensive experiments on natural-scene and remote sensing benchmarks, real-world datasets, and controlled diverse-degradation settings demonstrate that the proposed framework achieves superior quantitative performance, visual quality, and spectral fidelity compared with existing methods. These results validate the effectiveness of coordinated SVS prompting for robust HSI enhancement. Future work will focus on improving computational efficiency and extending the framework to more challenging deployment scenarios, such as real-time hyperspectral video processing, large-scale Earth observation, and onboard intelligent inference.

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